

MATH 13–Skeletal Notes

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October 10, 2026

Preface

These are **skeletal** notes. Every section carries the definitions, the statements of the theorems, and the outline of each example and each proof — but the details are left blank, and we fill them in together in class. What you are holding is the scaffolding of the course; the mathematics gets written into it as we do it.

So bring them with you. Before each class, print the section we are about to cover, or download it and annotate it on a tablet — the “PDF” button in the navigation bar gives you the whole book as one file to print from. Then complete the blanks in class as we work through the material. A page you filled in yourself is worth more later than any set of notes handed to you finished.

Everything for this course lives here: the notes themselves, the assignments, the review problems for each exam, and the solutions. Each item is added as we reach it, so check back regularly rather than assuming a page is final. Solutions to an assignment are posted only after its due date has passed; the review sets are posted the same way.

The problems are all gathered in the Exercises chapter. There are ten assignments, which are graded, and four sets of review problems, which are not — the review sets collect everything the assignments left over, and they are fair game on an exam. Assignments are due at 11:59 PM on the day shown below.

Assignments	Due day, at 11:59 PM
1, 2, 4, 5, 7, 8, 10	Friday
3, 6, 9	Wednesday (exam weeks 3, 6, and 9)

You hand your work in through Gradescope, not through this site. The steps — how to upload a scan or a photo of your written solutions, and how to tell Gradescope which page holds which problem — are under [Submitting Your Assignments in Gradescope](#) in the course syllabus. Read them once before the first assignment is due; a submission with the pages mismatched is the most common way points get lost for reasons that have nothing to do with the mathematics.

One more thing worth knowing before you start: each of the ten assignment pages carries a Socratic AI tutor that will coach you when you are stuck, without ever giving an answer away. How it works, what it will and will not do, and what to keep out of the chat are all set out under Using the AI Tutor, at the top of the Exercises chapter.

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Chapter 1

Hyperbolic Functions

Objectives

After completing this section, you should be able to:

- Define the hyperbolic sine and cosine in terms of e^x and e^{-x} , and differentiate them.
- Sketch the graphs of $\sinh x$ and $\cosh x$, and state their domains and ranges.
- Use the identity $\cosh^2 x - \sinh^2 x = 1$ and relate it to the parametrization of the hyperbola $x^2 - y^2 = 1$.
- Work with the remaining hyperbolic functions $\tanh x$, $\coth x$, $\operatorname{sech} x$, and $\operatorname{csch} x$, and apply hyperbolic identities to integrals and equations.

This chapter is about certain combinations of e^x and e^{-x} which appear in multiple branches of science and engineering. In many ways they are analogous to the trigonometric functions, and they have a similar relationship to the hyperbola that the trigonometric functions have to the circle, which is why they are known as **hyperbolic functions**. We will study these relationships gradually during this chapter.

These are skeletal notes: the definitions, identities, and problem statements are given, and blank boxes and empty grids are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

1.1 Definitions and Derivatives

The **hyperbolic cosine** is defined as

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad (1.1.1)$$

and the **hyperbolic sine** is defined as

$$\sinh x = \frac{e^x - e^{-x}}{2}. \quad (1.1.2)$$

Find the derivatives $\frac{d}{dx}(\cosh x)$ and $\frac{d}{dx}(\sinh x)$ by differentiating the two definitions.

workspace

1.2 Graph, Domain, and Range of $\cosh x$ and $\sinh x$

Since $\sinh x$ and $\cosh x$ are built from the two exponentials $\frac{1}{2}e^x$ and $\frac{1}{2}e^{-x}$, their graphs are easiest to understand by sketching those exponentials first. See [Figure 1.2.1](#) and [Figure 1.2.2](#).

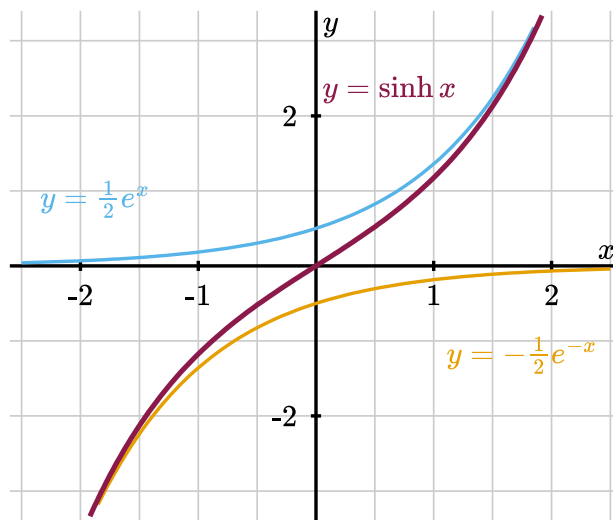


Figure 1.2.1 The graph of $y = \sinh x$. The curve is squeezed between the two exponentials $y = \frac{1}{2}e^x$ and $y = -\frac{1}{2}e^{-x}$, approaching the first as $x \rightarrow \infty$ and the second as $x \rightarrow -\infty$.

From the graph, read off the domain, the range, and the symmetry of $\sinh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\sinh x$ hugs $\frac{1}{2}e^x$ when x is large and positive, and hugs $-\frac{1}{2}e^{-x}$ when x is large and negative. Confirm this observation.

workspace

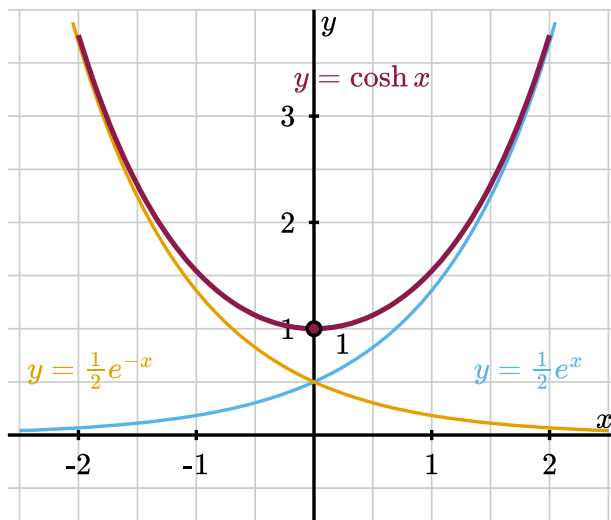


Figure 1.2.2 The graph of $y = \cosh x$. The curve is the sum of the two exponentials $y = \frac{1}{2}e^x$ and $y = \frac{1}{2}e^{-x}$, so it lies above both and has its minimum value 1 at $x = 0$.

From the graph, read off the domain, the range, and the symmetry of $\cosh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\cosh x$ hugs $\frac{1}{2}e^x$ when x is large and positive, and hugs $\frac{1}{2}e^{-x}$ when x is large and negative, while never dropping below 1. Confirm these observations.

workspace

1.3 Identities and Other Hyperbolic Functions

A similar identity to the trigonometric identity $\cos^2 x + \sin^2 x = 1$ holds for the hyperbolic functions:

$$\cosh^2 x - \sinh^2 x = 1. \quad (1.3.1)$$

As you already know, any point $P(x, y)$ on the circumference of the unit circle ($x^2 + y^2 = 1$) can be described in terms of sine and cosine of an angle $t \in [0, 2\pi)$, i.e. $x = \cos t$ and $y = \sin t$, which results in the trigonometric identity $\cos^2 t + \sin^2 t = 1$. Mark this on the circle in [Figure 1.3.1](#).

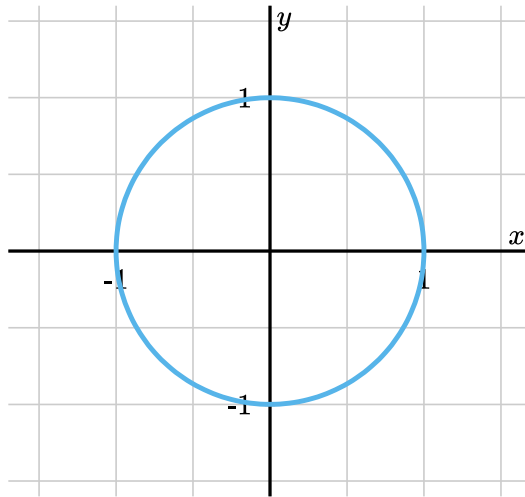


Figure 1.3.1 The unit circle $x^2 + y^2 = 1$. Mark a point $P(\cos t, \sin t)$ on it, and draw the right triangle with legs x and y and hypotenuse 1.

Similarly, any point $P(x, y)$ on the right branch of the hyperbola $x^2 - y^2 = 1$ can be represented as $x = \cosh t$ and $y = \sinh t$, where $t \in \mathbb{R}$. This follows directly from identity (1.3.1), i.e. $\cosh^2 t - \sinh^2 t = 1$. Mark this on the hyperbola in [Figure 1.3.2](#).

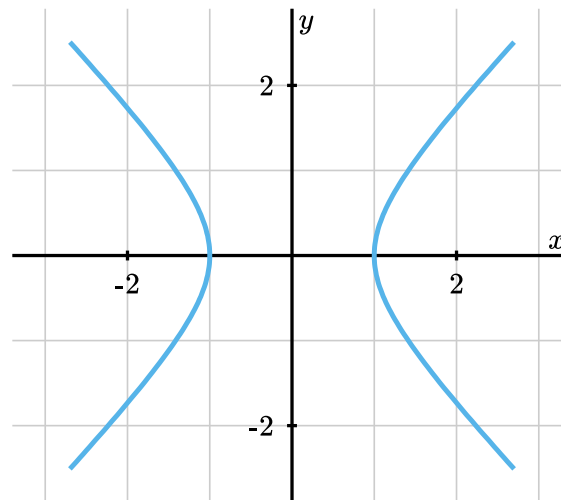


Figure 1.3.2 The hyperbola $x^2 - y^2 = 1$. Mark a point $P(\cosh t, \sinh t)$ on the right branch, and draw the segments from the origin to P and to the vertex.

Checkpoint 1.3.3 Parametrizing the Left Branch. Both branches of $x^2 - y^2 = 1$ appear in Figure 1.3.2, but the parametrization $P(\cosh t, \sinh t)$ traces only the right one, since $\cosh t \geq 1$ for every $t \in \mathbb{R}$. How would you parametrize the *left* branch, where $x \leq -1$?

Your answer and reasoning.

workspace

Hint. The identity $\cosh^2 t - \sinh^2 t = 1$ is unaffected if you change the sign of the first coordinate.

Checkpoint 1.3.4 True or False. The parametrization $x = -\cosh t$, $y = -\sinh t$, $t \in \mathbb{R}$, also traces the left branch of $x^2 - y^2 = 1$.

True or false? Your answer and reasoning.

workspace

As you might have guessed, the rest of the hyperbolic functions are defined

as follows.

$$\begin{aligned}\tanh x &= \frac{\sinh x}{\cosh x} & \operatorname{sech} x &= \frac{1}{\cosh x} \\ \coth x &= \frac{\cosh x}{\sinh x} & \operatorname{csch} x &= \frac{1}{\sinh x}\end{aligned}$$

1.4 Hyperbolic Identities

Below are some identities that you may find useful in some problems, however, you are not expected to memorize them. We will prove some of them as an exercise later. Each hyperbolic identity is listed next to the trigonometric identity it resembles.

Table 1.4.1 Hyperbolic identities and their trigonometric counterparts.

Hyperbolic	Trigonometric
$\cosh^2 x - \sinh^2 x = 1$	$\cos^2 x + \sin^2 x = 1$
$1 - \tanh^2 x = \operatorname{sech}^2 x$	$1 + \tan^2 x = \sec^2 x$
$\sinh 2x = 2 \sinh x \cosh x$	$\sin 2x = 2 \sin x \cos x$
$\cosh 2x = \cosh^2 x + \sinh^2 x$	$\cos 2x = \cos^2 x - \sin^2 x$
$\cosh^2 x = \frac{\cosh 2x + 1}{2}$	$\cos^2 x = \frac{1 + \cos 2x}{2}$
$\sinh^2 x = \frac{\cosh 2x - 1}{2}$	$\sin^2 x = \frac{1 - \cos 2x}{2}$

Notice the pattern: each hyperbolic identity is its trigonometric counterpart with the sign changed wherever two sines (or two tangents) are multiplied together.

Example 1.4.2 Computing $\cosh^{-1}(x)$. Show that the inverse hyperbolic cosine can be written in terms of the natural logarithm as

$$\cosh^{-1}(x) = \ln \left(x + \sqrt{x^2 - 1} \right), \quad x \geq 1.$$

Solution. (Restrict $\cosh$ to $x \geq 0$ so that it is one-to-one, set $y = \cosh(x)$, and solve for x .)

workspace



Example 1.4.3 An integral via a hyperbolic substitution. Use hyperbolic functions to calculate the integral

$$\int \frac{\sqrt{x^2 - 1}}{x^2} dx, \quad x \geq 1.$$

Hint: Similar to $\int \sec^2(x) dx = \tan(x) + C$, we have $\int \operatorname{sech}^2(x) dx = \tanh(x) + C$.

Solution.

workspace



1.5 Hyperbolic Functions in Real Life

Hyperbolic functions are more than a formal analogue of the trigonometric functions: they describe a number of concrete physical situations. We look at three of them.

1.5.1 *Hanging Cables and the Catenary

Starred section. This one is for the interested reader. It will not be examined.

If a heavy flexible cable (such as a telephone line, a power line, or a chain) hangs freely from two supports, it settles into a curve called a **catenary**. Contrary to a common guess, this curve is not a parabola; it is the graph of a hyperbolic cosine,

$$y = a \cosh\left(\frac{x}{a}\right),$$

where x is measured horizontally from the lowest point of the cable, so that the y -axis is the axis of symmetry and the lowest point sits at height $y(0) = a$. Sliding the curve up or down, as in $y = c + a \cosh(x/a)$, only changes where we draw the x -axis.

The shape comes out of a balance of forces. Look at the piece of cable running from the lowest point to a point where the cable makes an angle θ with the horizontal, and let s be the arc length of that piece. Three forces act on it: the tension T_0 at the lowest point, which is horizontal; the tension T

along the cable at the other end; and the weight $\lambda g s$ of the piece, where λ is the mass per unit length; see Figure 1.5.1. Balancing the horizontal and the vertical components gives

$$\begin{aligned} T \cos \theta &= T_0 \\ T \sin \theta &= \lambda g s. \end{aligned}$$

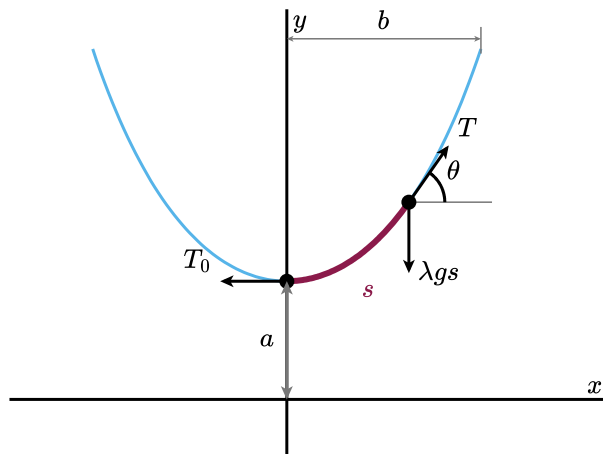


Figure 1.5.1 The three forces on the piece of cable of arc length s running from the lowest point of the cable to a point where the cable makes an angle θ with the horizontal. The lowest point sits at height $a = T_0/(\lambda g)$, and the supports are a distance $2b$ apart. (After Fig. 1 of Behroozi, cited below.)

The quantity $a = T_0/(\lambda g)$ has units of length, and dividing the second equation by the first eliminates T and leaves $s = a \tan \theta$. Since the cable makes the angle θ with the horizontal, $\tan \theta$ is its slope, so

$$\frac{dy}{dx} = \tan \theta = \frac{s}{a}.$$

Now use the arc length element $ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + (s/a)^2} dx$ and separate the variables:

$$dx = \frac{a ds}{\sqrt{a^2 + s^2}} \quad \implies \quad x = a \sinh^{-1} \left(\frac{s}{a} \right),$$

where the constant of integration vanishes because $s = 0$ at $x = 0$.

Solving for s gives $s = a \sinh(x/a)$, and therefore

$$\frac{dy}{dx} = \frac{s}{a} = \sinh \left(\frac{x}{a} \right).$$

One last integration, together with $y(0) = a$, produces the catenary equation $y = a \cosh(x/a)$.

Two things are worth noticing. First, a is a pure scale factor: written as $y/a = \cosh(x/a)$, the equation shows that every catenary is a scaled copy of the single curve $y = \cosh x$, in exactly the same way that every circle is a scaled copy of the unit circle; Figure 1.5.2 shows four of them. Second, a is fixed by the cable itself. If the cable has half-length l and its two supports are a distance $2b$ apart, then putting $s = l$ at $x = b$ in $s = a \sinh(x/a)$ gives

$$l = a \sinh \left(\frac{b}{a} \right),$$

which determines a (numerically) from the two lengths. Since $a = T_0/(\lambda g)$, a cable pulled tight has a large a and hangs almost flat, while a slack one has a small a and sags sharply.

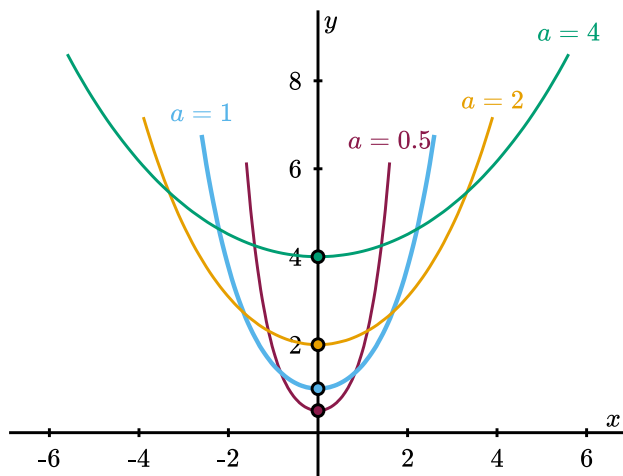


Figure 1.5.2 The catenaries $y = a \cosh(x/a)$ for $a = 0.5, 1, 2, 4$. Each curve meets the y -axis at its own value of a , and all four are scaled copies of the single curve $y = \cosh x$. (After Fig. 2 of Behroozi, cited below.)

A worked example of a hanging cable, in which we find the slope of the cable and the angle at which it meets its pole, appears in [Example 1.6.1](#).

The derivation above follows F. Behroozi, [In Praise of the Catenary](#), *The Physics Teacher* **56**, 214–217 (2018), which also discusses the sense in which all catenaries are similar to one another and suggests simple classroom demonstrations.

1.5.2 Celestial Mechanics

If a comet has enough speed, it can escape the gravitational pull of the sun, in which case one possible trajectory is a hyperbolic trajectory. The comet [2I/Borisov](#), discovered in 2019, is the first comet known to have come from outside our solar system. It was moving too fast for the sun to capture it, so its path is a hyperbola rather than an ellipse: it swung around the sun once and is now on its way back out. See [Figure 1.5.3](#).

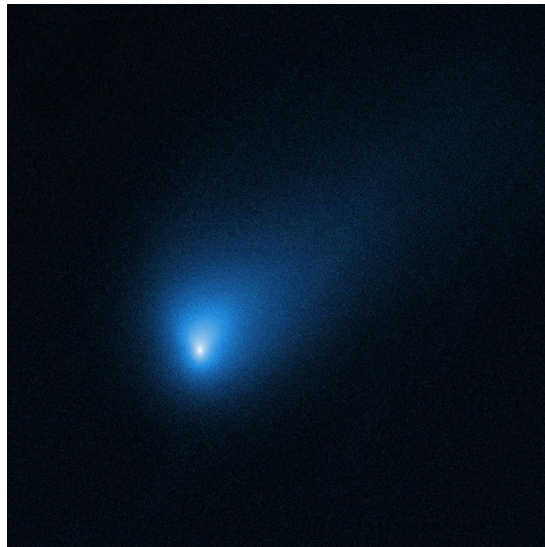


Figure 1.5.3 The interstellar comet 2I/Borisov, photographed by the Hubble Space Telescope in 2019. Because its speed exceeds the escape speed of the sun, its trajectory is a hyperbola and it passes through the solar system only once. (Image: NASA, ESA and D. Jewitt (UCLA).)

1.5.3 The Gateway Arch

The Gateway Arch in St. Louis, Missouri (designed in 1963 and completed in 1965) is a catenary turned upside down: flipping the curve converts the tension carried by a hanging chain into pure compression, which is what masonry and steel carry best. The geometric form of the gateway was set by Hannskari Bandel (structural engineer) and was expressed in the blueprints by the equation

$$y = A \left(\cosh \frac{Cx}{L} - 1 \right),$$

where A , C , and L are constants. The arch is slightly flattened compared with a uniform hanging chain, because it is thicker at the base than at the top. It stands 630 feet tall and 630 feet wide at the base; the National Park Service describes its construction and its geometry at [Gateway Arch National Park](#). See [Figure 1.5.4](#).



Figure 1.5.4 The Gateway Arch in St. Louis, Missouri. Its centerline follows the curve $y = A(\cosh(Cx/L) - 1)$, an upside-down catenary. (Photograph by John Margolies, 1988; John Margolies Roadside America photograph archive, Library of Congress, Prints and Photographs Division.)

1.6 Further Examples

Example 1.6.1 A hanging telephone line. A telephone line hangs between two poles 14 m apart in the shape of the catenary $y = 20 \cosh(x/20) - 15$, where x and y are measured in meters.

- A. Find the slope of this curve where it meets the right pole.
- B. Find the angle θ between the line and the pole.

Solution. (A sketch of the line between the poles at $x = -7$ and $x = 7$ may help.)

workspace

Example 1.6.2 Rewriting $\sinh(\ln(x))$. Consider the function $f(x) = \sinh(\ln(x))$. ▲

A. Express $f(x)$ as a fraction of two polynomials.

B. Calculate $\frac{df}{dx}$.

Solution.

workspace

▲

Example 1.6.3 A double-angle identity. Prove that $\sinh(2x) = 2 \sinh(x) \cosh(x)$.

Proof (start from the definitions).

workspace



Example 1.6.4 Solving a hyperbolic equation. Consider the equation $2 \cosh(2x) + 10 \sinh(2x) = 5$ and solve for x .

Solution.

workspace



Chapter 2

Series and Sequences

2.1 Infinite Series

Objectives

After completing this section, you should be able to:

- Define an infinite sequence, an infinite series, and the sequence of partial sums, and decide convergence of a series from its partial sums.
- Recognize a geometric series, and determine whether it converges and what it converges to.
- Apply the n th Term Test for divergence, and understand what the test can and cannot conclude.
- Combine convergent series using the sum, difference, and constant multiple rules, and evaluate telescoping series from their partial sums.

What does it mean to add infinitely many numbers together? In this section we answer that question by defining an infinite series and its sequence of partial sums, so that convergence of a series becomes a statement about a limit we already know how to study. We then put the definition to work on the geometric series, whose behavior we can settle completely, and on the n th Term Test, which is our first quick check for divergence.

These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

2.1.1 Definitions

Definition 2.1.1 Sequence. A **sequence** is a list of numbers, $a_1, a_2, a_3, \dots$. An **infinite sequence** of numbers is a function whose domain is the set of positive integers. ◆

Example 2.1.2 A Sequence of Halves. The numbers $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$ form an infinite sequence; its n th term is $a_n = \left(\frac{1}{2}\right)^n$. ▲

Definition 2.1.3 Infinite Series. The sum of the numbers in an infinite

sequence $\{a_n\}$, i.e. $a_1 + a_2 + a_3 + \cdots = \sum_{n=1}^{\infty} a_n$, is called an **infinite series**.

Here a_n is the n th term of the series. ◆

Example 2.1.4 An Infinite Series with a Finite Sum. Infinite sequences can have finite sums. Consider the sum of the sequence from [Example 2.1.2](#), i.e. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$.

It is most convenient to evaluate the result of this sum geometrically. Draw a square of side one in the space below. Shade half of it, then half of what is left, then half of what is left after that, and keep going. Label the pieces $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$. What is the total shaded area, and therefore what is the sum?

draw the unit square and its subdivisions here



2.1.2 Partial Sums

Consider again the infinite sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$. Let us denote the sum of the first n terms in this sequence by s_n , which are known as **partial sums**. Can we find a pattern in the sequence of partial sums?

Write out s_1 , s_2 , and s_3 , each one both as a fraction and in the form $1 - \square$.

Then guess a closed formula for $s_n = \sum_{k=1}^n \left(\frac{1}{2}\right)^k$.

$$s_1 =$$

$$s_2 =$$

$$s_3 =$$

$$\vdots$$

$$s_n =$$

Now use your formula for s_n to compute the infinite series, by taking the limit of the partial sum as $n \rightarrow \infty$. Compare the answer with the one you found geometrically from the square.

workspace

In the example above, the sequence of partial sums converged to a number. In general, if the sequence of the partial sums converges to a number, we say that the series **converges**, otherwise we say that the series **diverges**.

2.1.3 Geometric Series

An important example of infinite series is the geometric series. The **geometric series** is of the form

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}, \quad (2.1.1)$$

where a, r are real numbers and $a \neq 0$. Note that we can re-write the series as

$$\sum_{n=0}^{\infty} ar^n.$$

Let us compute the partial sum s_n for the geometric series, so that

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1}. \quad (2.1.2)$$

Multiply (2.1.2) by r , subtract the result from (2.1.2), and solve for s_n . Almost every term should cancel.

$$r s_n =$$

$$s_n - r s_n =$$

$$s_n =$$

Next, take the limit of your partial sum as $n \rightarrow \infty$. Treat the cases $|r| < 1$, $|r| > 1$, and $r = 1$ separately, and in each case say what $\lim_{n \rightarrow \infty} r^n$ does.

$$|r| < 1:$$

$$|r| > 1:$$

$$r = 1:$$

Checkpoint 2.1.5 What Happens When $r = -1$? We have now handled $|r| < 1$, $|r| > 1$, and $r = 1$, but one case is still missing: $r = -1$. Write out the partial sums s_1, s_2, s_3, s_4 of the series $a - a + a - a + \cdots = \sum_{n=1}^{\infty} a(-1)^{n-1}$ for $a \neq 0$. Does the sequence of partial sums approach a single number as $n \rightarrow \infty$? What does that tell you about the series?

Your answer and reasoning.

workspace

Hint. The partial sums do not grow without bound here, the way they do

when $r = 1$. Look instead at whether they settle down to one value.

Putting all of the cases together, we can say that if $|r| \geq 1$, the geometric series is divergent.

Summary.

The geometric series (2.1.1) converges to $\frac{a}{1-r}$ if $|r| < 1$, i.e.

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1, \quad (2.1.3)$$

and diverges if $|r| \geq 1$.

Next, as an application of the geometric series, we will go through the following example, which is from our textbook.

Example 2.1.6 A bouncing ball. You drop a ball from a meters above a flat surface. Each time the ball hits the surface after falling a distance h , it rebounds a distance rh , where r is positive but less than 1. Find the total distance the ball travels up and down.

Solution. (Use Figure 2.1.7 to write the total distance as an infinite series, then match it against (2.1.3). Watch the first drop: it is travelled only once.)

workspace



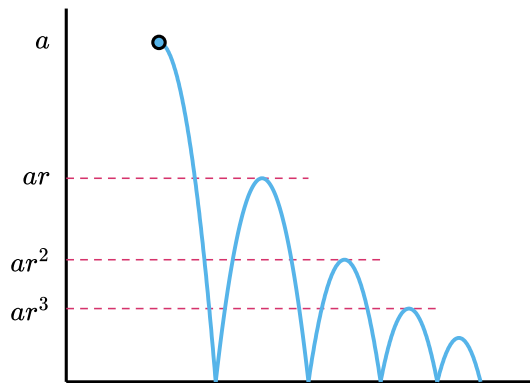


Figure 2.1.7 The ball falls a distance a , then rises and falls a distance ar , then ar^2 , and so on, so the total distance travelled is $a + 2ar + 2ar^2 + \cdots$.

2.1.4 The n th Term Test

Theorem 2.1.8 If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Important note: If $\lim_{n \rightarrow \infty} a_n = 0$, we *cannot* conclude that $\sum_{n=1}^{\infty} a_n$ converges. See parts D and E in the example below. Also see [Example 2.1.12](#).

The following test is a consequence of the above theorem.

The n th Term Test.

If $\lim_{n \rightarrow \infty} a_n$ does not exist or $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Example 2.1.9 Testing series for convergence. Determine whether the series is convergent or divergent. If it is convergent, find its sum.

A. $\sum_{n=1}^{\infty} n$

B. $\sum_{n=1}^{\infty} (-1)^n$

C. $\sum_{n=1}^{\infty} \frac{n^3 + 1}{n^2 + n}$

D. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

E. $\frac{4}{9} - \frac{8}{27} + \frac{16}{81} - \frac{32}{243} + \cdots$

Solution.

A.

B.

C.

D.

E.



2.1.5 Combining Series

Theorem 2.1.10 If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. Sum Rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. Difference Rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. Constant Multiple Rule: $\sum ka_n = k \sum a_n = kA$ (Any number k).

Example 2.1.11 Using the difference rule. Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{2^n}$.

Solution. (You have already summed both of these series in this section: one in [Subsection 2.1.2](#) and the other in part D of [Example 2.1.9](#).)

workspace



Example 2.1.12 A telescoping series with logarithms. Evaluate $\sum_{k=0}^{\infty} \ln\left(\frac{k+2}{k+1}\right)$, if it converges and if it diverges, show that it does.

Solution.

workspace



2.2 Taylor and Maclaurin Series

Objectives

After completing this section, you should be able to:

- Approximate a function near a point by its tangent line and by higher order polynomials built from its derivatives.
- Write down the Taylor series generated by f at $x = a$ and the Maclaurin series generated by f .
- Compute the Taylor polynomial of order n generated by f at $x = a$.
- Find Maclaurin series of standard functions such as $\sin(x)$ directly from the definition.

2.2.1 Definitions of Taylor Series, Maclaurin Series, and Taylor Polynomials

In this section we will answer the following question:

If all we know about a function is information at $x = a$, i.e. $f(a), f'(a), f''(a), \dots$, how can we approximate $f(x)$ with a polynomial $p(x)$?

These are skeletal notes: the definitions and problem statements are given, and blank boxes are left wherever a derivation or a solution belongs, so that you can fill them in by hand.

Throughout, let f be a function with derivatives of all orders throughout some interval containing a as an interior point.

Before stating the definitions, let us see where the coefficients of such a polynomial have to come from. Suppose the only things we know about f are its readings at $x = a$: the value $f(a)$, the slope $f'(a)$, the second derivative $f''(a)$, and so on. We look for a polynomial written in terms of $x - a$,

$$p_n(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + \dots + c_n(x - a)^n, \quad (2.2.1)$$

and we pin down the unknown coefficients $c_0, c_1, \dots, c_n$ by requiring p_n to agree with f at $x = a$ in as many derivatives as it has coefficients. Writing it in terms of $x - a$ rather than x is what makes this manageable: every term after the first vanishes at $x = a$.

Work out the first few coefficients. In each row, differentiate the polynomial as many times as it has coefficients, evaluate at $x = a$, and match against f .

a line, $p_1(x) = c_0 + c_1(x - a)$: $c_0 =$ $c_1 =$

a parabola, $p_2(x) = c_0 + c_1(x - a) + c_2(x - a)^2$: $c_2 =$

a cubic, $p_3(x) = \dots + c_3(x - a)^3$: $c_3 =$

Notice what should *not* have happened: c_0 and c_1 should come out the same every time. Each new condition is the first one in which the next coefficient appears, so it determines that coefficient and leaves the earlier ones untouched. That is why p_2 is the tangent line with a single new term added to it, rather than a fresh approximation built from scratch. The pattern in the denominators should now be visible: $0!, 1!, 2!, 3!, \dots$

One computation settles every coefficient at once. Differentiate (2.2.1) exactly k times and evaluate at $x = a$. Say what happens to the terms of degree below k , to those of degree above k , and to the term $c_k(x - a)^k$ itself; then impose $p_n^{(k)}(a) = f^{(k)}(a)$ and solve for c_k .

$$p_n^{(k)}(a) =$$

$$c_k =$$

So there is nothing to choose. Once we ask a polynomial to match f and its first n derivatives at $x = a$, its coefficients are determined, and they are the numbers $f^{(k)}(a)/k!$. Reading $f^{(0)}$ as f and $0!$ as 1, the first coefficient $c_0 = f(a)$ fits the same formula. Letting n grow without bound leads to the following definitions.

Definition 2.2.1 Taylor Series. The **Taylor series generated by f at $x = a$** is

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \cdots \\ &\quad + \frac{f^{(n)}(a)}{n!} (x-a)^n + \cdots \end{aligned}$$

◆

Definition 2.2.2 Taylor Polynomial of Order n . The **Taylor polynomial of order n** generated by f at $x = a$ is the polynomial

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

◆

Remark 2.2.3 Maclaurin Series. The centre $a = 0$ is common enough to have its own name. The Taylor series generated by f at $x = 0$ is known as the **Maclaurin series generated by f** , which is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n + \cdots.$$

Example 2.2.4 Approximating e^x near $x = 0$. Write down the Taylor polynomials of orders 1, 2, and 3 generated by $f(x) = e^x$ at $x = 0$.

Solution. Every derivative of e^x is e^x again, so the coefficients of Definition 2.2.2 are easy. Check your answers against Figure 2.2.5.

$$p_1(x) =$$

$$p_2(x) =$$

$$p_3(x) =$$

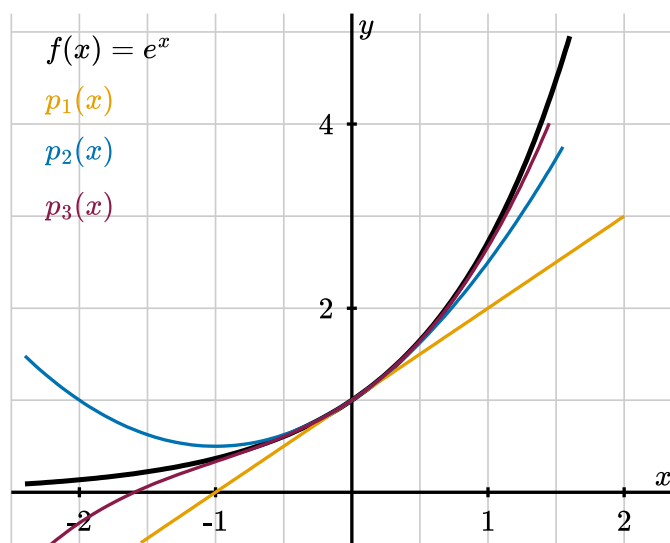


Figure 2.2.5 The function $f(x) = e^x$ together with the Taylor polynomials p_1 , p_2 , and p_3 at $x = 0$. Near $x = 0$ each polynomial hugs the curve more closely than the one before it.

An animation of this construction accompanies the online version of these notes, with sliders for the centre a and the order n . Two things are worth noting from it. Moving a rebuilds the same construction at a new point of the curve, where p_n again touches f . Raising n adds exactly one term, $\frac{f^{(n)}(a)}{n!}(x - a)^n$, which vanishes at $x = a$ — so it changes the shape of the polynomial everywhere else but never moves it off the point $(a, f(a))$.

The interval on which p_n stays within 0.1 of f widens with every term: about $[-0.5, 0.4]$ for p_1 , about $[-1.3, 1.2]$ for p_3 , and wider still by p_8 . So the Taylor polynomials do not merely improve at the centre; the region where they are usable grows. Whether that region eventually covers everything, and how large the error is at a given x , are the questions of the next section.

2.2.2 More Examples

Example 2.2.6 The Maclaurin series of $\sin(x)$. Consider the function $f(x) = \sin(x)$.

- A. Find the Taylor series generated by f at $x = 0$. Note that this is the same as the Maclaurin series generated by f .

- B. Calculate the first four Taylor polynomials $p_n(x)$.
- C. Plot the original function $f(x) = \sin(x)$ and the Taylor polynomials obtained in part B to confirm that the higher order polynomials provide a better approximation.
- A. Differentiate f repeatedly, evaluate at $x = 0$, find the pattern, then use [Remark 2.2.3](#).

derivatives:

at $x = 0$:

pattern:

series:

- B. Keep the first four non-zero polynomials.

$$p_1(x) =$$

$$p_3(x) =$$

$$p_5(x) =$$

$$p_7(x) =$$

- C. Check your polynomials against [Figure 2.2.7](#).



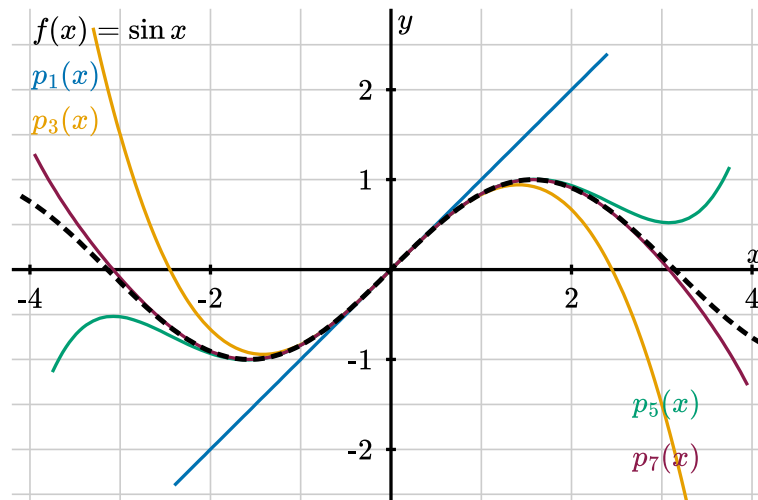


Figure 2.2.7 The function $f(x) = \sin x$ together with the Taylor polynomials at $x = 0$, where $p_1(x) = x$, $p_3(x) = x - \frac{x^3}{3!}$, $p_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$, and $p_7(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$.

Checkpoint 2.2.8 Consider the function $f(x) = \frac{x}{1 - 9x}$.

- Compute the Maclaurin series generated by f . Express the result in sigma notation.
- Find the interval of convergence for this series.

A.

B.

Hint. Write $\frac{1}{1 - 9x}$ as a geometric series first.

Checkpoint 2.2.9 Compute the Taylor series of $f(x) = \frac{1}{3x + 7}$ at $x = 1$.

workspace

Hint. Either differentiate repeatedly at $x = 1$, or write $3x + 7$ in terms of $x - 1$ and reduce to a geometric series.

A list of Taylor series. The Maclaurin series below are derived or used throughout this book, and the first four are the ones on the formula sheet of the exams. Each of them is a Taylor series centered at $a = 0$, so each can be obtained from [Definition 2.2.1](#) by computing derivatives, as we did for e^x and $\sin(x)$. Most of the time, though, it is faster to start from one of these and substitute, differentiate, integrate, or multiply: that is how the series for $\cos(2x)$, e^{-x^2} , $\frac{\sin x}{x}$, and $\arctan(x)$ are found in this book.

Table 2.2.10 A list of Taylor series: the Maclaurin series derived or used in this book.

Series	Interval of convergence	Where in the book
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ $= 1 + x + x^2 + x^3 + \cdots$	$(-1, 1)$	The geometric series, Subsection 2.1.3 . Replacing x by $-x$, x^2 , or $-x^2$ gives the series
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ $= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$(-\infty, \infty)$	Derived in Example 2.2.4 . Convergence proved in Subsection 2.3.2 .
$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ $= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$	$(-\infty, \infty)$	Derived in Example 2.2.6 . Divided by x in Example 2.3.13 .
$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$ $= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$	$(-\infty, \infty)$	The derivative of the series of $\sin(x)$. Used, with $2x$ and $5x$ in place of x , in Assignment
$\sinh(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$ $= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$	$(-\infty, \infty)$	Derived in Sample Past Exam 1; it is the odd
$\cosh(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$ $= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$	$(-\infty, \infty)$	The derivative of the series of $\sinh(x)$; it is the
$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ $= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$(-1, 1]$	Its Taylor polynomials, for $\ln(1+2x)$ and $\ln(1$
$\arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$ $= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$	$[-1, 1]$	Obtained by integrating the series of $\frac{1}{1+t^2}$ in S Evaluated at $x = 1$ to approximate π in Exam

Two more series in this book come from the table by substitution, and both converge on $(-\infty, \infty)$: $e^{-x^2} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!}$, used in [Example 2.3.12](#) to approximate an integral, and $\frac{\sin x}{x} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!}$, used in [Example 2.3.13](#) to evaluate one exactly.

2.3 Taylor Series and its convergence

Objectives

The main objectives of this section are to answer the following questions:

1. Under what conditions is the function $f(x)$ equal to the Taylor series generated by f at $x = a$? That is, when can we say $f(x) =$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n?$$

2. If the Taylor polynomial of degree n , $p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$, is used as an approximation to $f(x)$, how can we estimate the error in this approximation? That is, how can we estimate the magnitude of $R_n(x) = f(x) - p_n(x)$?

[Theorem 2.3.1](#) and [Theorem 2.3.4](#) will enable us to answer these questions. We begin by stating Taylor's formula and the remainder estimation theorem, and then use them to bound the error in a Taylor approximation, discuss some applications of these theorems, and close with the proof of the remainder theorem.

These are skeletal notes: the theorems and problem statements are given, and blank boxes are left wherever a derivation or a solution belongs, so that you can fill them in by hand.

2.3.1 Taylor's Formula and The Remainder Estimation Theorem

Theorem 2.3.1 Taylor's Formula. *Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a . Then for each $x \in I$ and for each positive integer n , there exists a number c between a and x such that*

$$f(x) = p_n(x) + R_n(x), \quad (2.3.1)$$

where

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k, \quad (2.3.2)$$

and

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}. \quad (2.3.3)$$

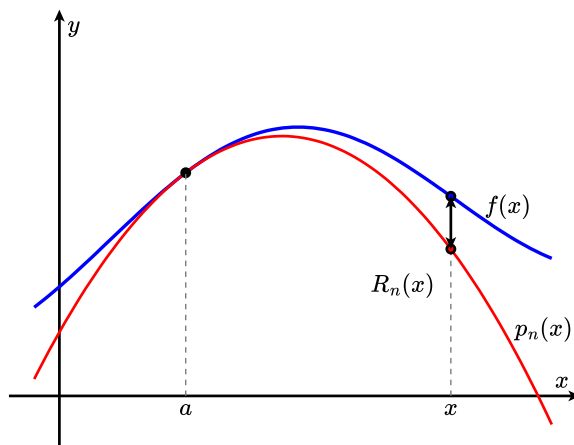


Figure 2.3.2 Taylor's Formula: $f(x) = p_n(x) + R_n(x)$. The Taylor polynomial p_n agrees with f at $x = a$; away from a , the vertical gap between them is the remainder $R_n(x)$.

Remark 2.3.3 Finding an upper bound for the error term without knowing the value of c . Usually the value of c is not explicitly known. However, we may manage to find *an upper bound* for the error term $R_n(x)$

without knowing the exact value of c . This is achieved by finding an upper bound M for $|f^{(n+1)}(c)|$, where c is between a and x , and then using this upper bound to estimate the error term.

This is the idea behind every error estimate in this section. It is used in [Subsection 2.3.2](#) to prove that the Taylor series of e^x converges to e^x for every x , and in each of the three examples of [Subsection 2.3.3](#).

Theorem 2.3.4 The Remainder Estimation Theorem. *Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a . Then for each $x \in I$ and for each positive integer n , there exists a number c between a and x such that*

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}, \quad (2.3.4)$$

where M is an upper bound for $|f^{(n+1)}(c)|$ on the interval between a and x .

(Derive (2.3.4) from (2.3.3).)

workspace

You are not expected to know the proof of Taylor's formula itself, only to understand the statement and its implications. The proof is worked through in [Subsection 2.3.5](#).

2.3.2 An example of a Taylor series that converges

Consider the function $f(x) = e^x$. The Taylor series generated by f at $x = 0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \underline{\hspace{2cm}}, \quad (2.3.5)$$

and its remainder is

$$R_n(x) = \underline{\hspace{2cm}}. \quad (2.3.6)$$

(Compute (2.3.5) and (2.3.6).)

Taylor series of e^x at $x = 0$:

$R_n(x) =$

(Show that (2.3.5) converges to e^x for every $x \in \mathbb{R}$.)

workspace

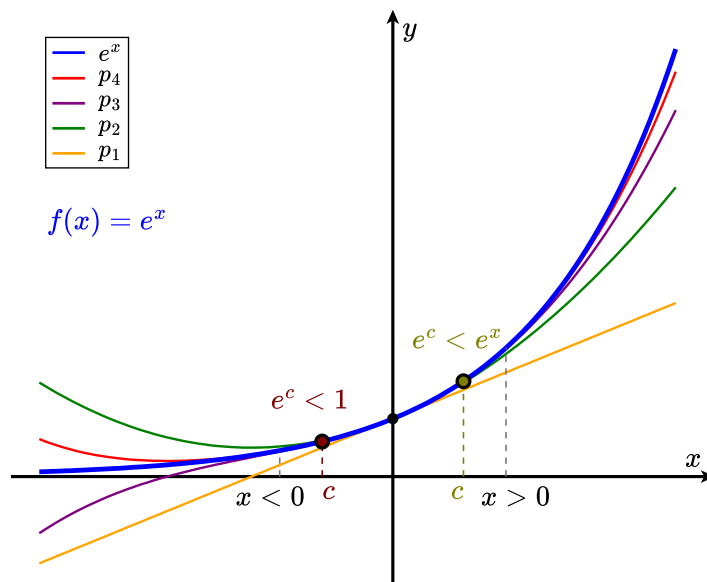


Figure 2.3.5 Schematic graph of $f(x) = e^x$ and its Taylor polynomials $p_1, \dots, p_4$ about 0.

An animation of this convergence — the Taylor polynomials p_n hugging the graph of $f(x) = e^x$ over a wider and wider interval as n grows — accompanies the online version of these notes.

2.3.3 The error in using a Taylor polynomial

Suppose we approximate $f(x)$ by the Taylor polynomial (2.3.2) of degree n at $x = a$. The error is $R_n(x) = f(x) - p_n(x)$, which is (2.3.1) rearranged, and is given by (2.3.3). To bound it we need an upper bound M for $|f^{(n+1)}(c)|$ on the interval between a and x , as in Remark 2.3.3. The three examples below carry out that step in three different settings.

Example 2.3.6 Approximating e^x using a Taylor polynomial. Approximate $e^{0.5}$ using the Taylor polynomial of degree 2 at $x = 0$. Find an upper bound for the error in this approximation.

Solution.

$$p_2(x) =$$

bound on $|R_2(0.5)|$:

actual error:



Example 2.3.7 Given a desired error bound, find values of x for which the approximation is guaranteed to be valid. Assume that we use the Taylor polynomial of degree 3 at $x = 0$ to approximate $\sin(x)$. For approximately what values of x can you replace $\sin(x)$ by such a Taylor polynomial with an error of magnitude no greater than 4×10^{-3} ?

Solution.

$$p_3(x) = \quad \text{and} \quad R_4(x) =$$

bound on $|R_4(x)|$:

solve for $|x|$:



Example 2.3.8 Finding the value of n such that the error is less than a given tolerance. Find the smallest value of n for which the polynomial approximation for $\sin(x)$ is accurate to ± 0.005 for values of x in the interval $[-\pi/2, \pi/2]$.

Solution.

bound on $|R_n(x)|$:

inequality to solve:

test values of n :



2.3.4 Applications of Taylor Series and Remainder Theorem

We begin with two uses of Taylor series that belong to single-variable calculus: computing limits of the indeterminate form $\frac{0}{0}$, which along the way gives a proof of L'Hôpital's rule, and approximating definite integrals whose integrands have no elementary antiderivative.

The remaining three parts of this subsection are starred. Physicists often use Taylor series to approximate functions in order to simplify calculations. The first example is the kinetic energy of an object in relativistic mechanics. The second example is the approximation of the period of a pendulum. We will discuss the error in using these approximations and when it is valid to use them. Additionally, we will discuss how to use the Taylor series to estimate the value of π in the last example.

2.3.4.1 Computing Limits and Integrals with Taylor Series

Taylor's formula [Theorem 2.3.1](#) replaces a function near a point by a polynomial plus a remainder whose size the Remainder Estimation Theorem controls. That is exactly what two problems from single-variable calculus call for: a limit of the indeterminate form $\frac{0}{0}$, where numerator and denominator both vanish and the question is *how fast*, and a definite integral whose integrand has no elementary antiderivative.

We will use two Maclaurin series. The series of $\sin(x)$ was found in [Example 2.2.6](#):

$$\begin{aligned}\sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \\ &= x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots, \quad x \in \mathbb{R}.\end{aligned}\tag{2.3.7}$$

The series of $\arctan(x)$ is

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots, \quad |x| \leq 1;\tag{2.3.8}$$

it comes from the geometric series [\(2.1.1\)](#): since $\frac{d}{dt} \arctan(t) = \frac{1}{1+t^2}$, and $\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \cdots$ for $|t| < 1$, integrating this from 0 to x term by term gives [\(2.3.8\)](#).

Example 2.3.9 A $\frac{0}{0}$ limit by Taylor series. Compute

$$\lim_{x \rightarrow 0} \frac{\arctan(x) - x}{\sin(x) - x}.$$

Solution.

numerator and denominator as series:

factor and cancel:

limit:

why the remaining terms vanish:



Example 2.3.10 Proving L'Hôpital's rule with Taylor's formula. Let f and g have continuous second derivatives on an open interval containing a , with $f(a) = g(a) = 0$ and $g'(a) \neq 0$. Use Taylor's formula to prove L'Hôpital's rule:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.\tag{2.3.9}$$

Solution.

Taylor's formula for f and for g :

after cancelling the common factor:

the limit as x tends to a :

the second equality:



Remark 2.3.11 Why the rule sometimes has to be applied several times. If $f'(a) = g'(a) = 0$ as well, the quotient $f'(x)/g'(x)$ is again of the form $\frac{0}{0}$ and L'Hôpital's rule has to be applied again. Taylor's formula explains what is going on. Suppose f and g have $k + 1$ continuous derivatives near a , their first $k - 1$ derivatives vanish at a , and $g^{(k)}(a) \neq 0$. Then Taylor's formula with $n = k$ gives

$$\begin{aligned} \frac{f(x)}{g(x)} &= \frac{\frac{f^{(k)}(a)}{k!}(x-a)^k + R_k(x)}{\frac{g^{(k)}(a)}{k!}(x-a)^k + \tilde{R}_k(x)} \\ &\longrightarrow \frac{f^{(k)}(a)}{g^{(k)}(a)} \quad \text{as } x \rightarrow a, \end{aligned}$$

because each remainder is at most a constant times $|x - a|^{k+1}$. The limit is the ratio of the first nonzero Taylor coefficients, which is exactly what k applications of L'Hôpital's rule compute. In [Example 2.3.9](#) the first nonzero coefficients were those of x^3 . Check that $f'''(0)/g'''(0)$, with $f(x) = \arctan(x) - x$ and $g(x) = \sin(x) - x$, agrees with the limit you found.

Example 2.3.12 A definite integral with no elementary antiderivative. The function e^{-x^2} has no elementary antiderivative, so the integral

$$I = \int_0^1 e^{-x^2} dx$$

cannot be evaluated with the fundamental theorem of calculus. Use the Taylor polynomial of order n of e^u to approximate I , bound the error with the Remainder Estimation Theorem, and find a value of n that guarantees an error of at most 10^{-4} .

Solution.

series with remainder, and the bound on the remainder:

integrate from 0 to 1:

error bound:

choice of n :



Example 2.3.13 A non-elementary integral computed exactly. In [Example 2.3.12](#), term-by-term integration produced a series that we could only *estimate*. Sometimes, however, the series produced by term-by-term integration is one we already recognize, and then the integral can be evaluated *exactly*, even though the integrand has no elementary antiderivative. Show that

$$\int_0^\infty e^{-x} \frac{\sin x}{x} dx = \frac{\pi}{4}.$$

Hint. Each power of x has to be integrated against e^{-x} on $[0, \infty)$. Use the following fact from Calculus II, proved by integration by parts¹: for every integer $k \geq 0$,

$$I_k = \int_0^\infty x^k e^{-x} dx = k!. \quad (2.3.10)$$

*Solution.*²

a series for the integrand:

integrate term by term:

recognize the series:



2.3.4.2 *Approximating Relativistic Kinetic Energy

Starred section. This one is for the interested reader. It will not be examined.

In relativistic mechanics, the mass of an object moving with velocity v is given by:

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (2.3.11)$$

where m_0 is the rest mass of the object and c is the speed of light. Then the kinetic energy of an object of mass m moving with velocity v is:

$$K = mc^2 - m_0c^2 \quad (2.3.12)$$

In the case when $v \ll c$, we can use the Taylor series to approximate the kinetic energy. In [Example 2.3.15](#) we will show that the kinetic energy can be

¹Let $I_k = \int_0^\infty x^k e^{-x} dx$. Then $I_0 = \int_0^\infty e^{-x} dx = 1$. For $k \geq 1$, integration by parts with $u = x^k$ and $dv = e^{-x} dx$ gives $I_k = \left[-x^k e^{-x} \right]_0^\infty + k I_{k-1} = k I_{k-1}$, since $x^k e^{-x} \rightarrow 0$ as $x \rightarrow \infty$. By induction, $I_k = k(k-1) \cdots 1 \cdot I_0 = k!$.

²Integrating an infinite series term by term over the unbounded interval $[0, \infty)$ needs a justification, and the Remainder Estimation Theorem supplies one. Let $S_N(x) = \sum_{n=0}^N \frac{(-1)^n x^{2n}}{(2n+1)!}$ be the partial sum of the series for $\frac{\sin x}{x}$, a polynomial of degree $2N$. For $f(x) = \sin x$ every derivative satisfies $|f^{(k)}(t)| \leq 1$. The Taylor polynomial of $\sin x$ of degree $2N+2$ has no x^{2N+2} term, so it equals $x S_N(x)$, and the Remainder Estimation Theorem gives $|\sin x - x S_N(x)| \leq \frac{|x|^{2N+3}}{(2N+3)!}$. Dividing by $x > 0$, $\left| \frac{\sin x}{x} - S_N(x) \right| \leq \frac{x^{2N+2}}{(2N+3)!}$. Multiplying by e^{-x} , integrating, and using $I_k = k!$ with $k = 2N+2$, the integral differs from the partial sum $\sum_{n=0}^N \frac{(-1)^n}{2n+1}$ by at most $\frac{(2N+2)!}{(2N+3)!} = \frac{1}{2N+3}$. This bound tends to 0 as $N \rightarrow \infty$, so the partial sums of the Leibniz series converge to the integral. Since they also converge to $\pi/4$, the integral equals $\pi/4$. A numerical evaluation of the integral gives 0.785398..., which agrees with $\pi/4 = 0.785398 \dots$. The bound $\frac{1}{2N+3}$ also shows why the Leibniz series converges slowly: to guarantee an error below 10^{-3} we need roughly 500 terms.

approximated by the formula $K \approx \frac{1}{2}m_0v^2$ when $v \ll c$. See Figure 2.3.14 for a comparison of the relativistic kinetic energy and its Newtonian approximation.

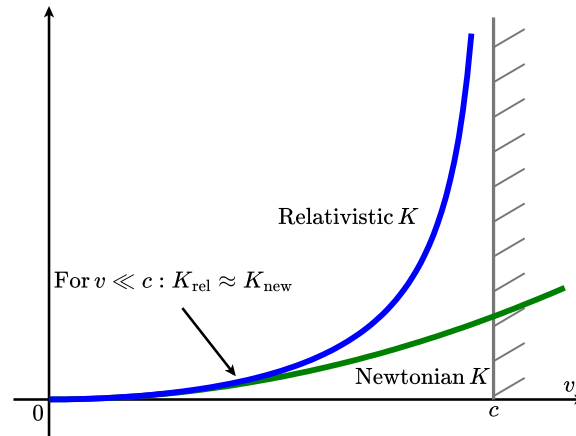


Figure 2.3.14 Relativistic versus Newtonian kinetic energy. The relativistic energy races toward a wall at the speed of light c , while the Newtonian energy follows a gentle parabola; the two are approximately in agreement only when $v \ll c$.

Example 2.3.15 Newtonian kinetic energy as an approximation to relativistic kinetic energy when $v \ll c$. Show that the kinetic energy of an object moving with velocity v can be approximated by the formula $K \approx \frac{1}{2}m_0v^2$ when $v \ll c$.

Solution.

workspace

When $v \ll c$, we can therefore approximate the kinetic energy as:

$$K \approx \frac{1}{2}m_0v^2. \quad (2.3.13)$$



Example 2.3.16 Estimating the error in using the Newtonian kinetic energy formula to approximate the relativistic kinetic energy. Assume

that a car is moving with a velocity of $v = 30m/s$ (≈ 67 miles per hour). Use the remainder's theorem to estimate the error in using the Newtonian kinetic energy formula (2.3.13) to approximate the relativistic kinetic energy of the car. The speed of light is $c = 3 \times 10^8 m/s$.

Solution.

workspace



2.3.4.3 *Small-Angle Approximation for a Pendulum

Starred section. This one is for the interested reader. It will not be examined.

We begin this section by briefly reviewing the forces acting on a simple pendulum and how the small-angle approximation allows us to treat its motion as simple harmonic motion.

The bob moves along the arc, so only the component of gravity *tangent* to that arc drives the motion. Resolving the weight $m\vec{g}$ into a component along the string ($mg \cos \theta$, balanced by the string tension $\vec{F}_T$) and a component tangent to the arc, as shown in Figure 2.3.17, gives the restoring force

$$F = -mg \sin \theta. \quad (2.3.14)$$

The minus sign indicates that the force always points back toward the equilibrium (straight-down) position. This is *not* Hooke's law: the force is proportional to $\sin \theta$, not to the displacement itself, so the motion is not exactly simple harmonic. Writing the arc displacement as $x = \ell\theta$, we would need F to be proportional to x —that is, to θ —for the motion to be simple harmonic.

The small-angle approximation bridges this gap. From the Taylor series

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots, \quad (2.3.15)$$

when θ is small (in radians) the higher-order terms are negligible and $\sin \theta \approx \theta$. The restoring force then becomes

$$F \approx -mg\theta = -\frac{mg}{\ell} x,$$

which *is* Hooke's law $F = -kx$ with effective spring constant $k = \frac{mg}{\ell}$.

This is exactly what fixes the period. Newton's second law $F = ma = m \frac{d^2x}{dt^2}$ turns Hooke's law into the equation of motion

$$m \frac{d^2x}{dt^2} = -\frac{mg}{\ell} x, \quad \text{that is,} \quad \frac{d^2x}{dt^2} = -\frac{g}{\ell} x.$$

Notice that the mass cancels. The equation says that x is a function whose second derivative is a negative multiple of itself, and the functions with that property are the sines and cosines: writing $\omega^2 = \frac{g}{\ell}$, every solution has the form

$$x(t) = A \cos(\omega t) + B \sin(\omega t), \quad \omega = \sqrt{\frac{g}{\ell}},$$

which you can verify by differentiating twice. The number ω is the angular frequency, and $\cos(\omega t)$ and $\sin(\omega t)$ repeat when ωt increases by 2π . So the motion repeats after a time T with $\omega T = 2\pi$, giving

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{\ell}{g}}. \quad (2.3.16)$$

Thus, for small swings the pendulum behaves as a simple harmonic oscillator, with the period (2.3.16), which is independent of both the amplitude and the mass. Taylor's Remainder Theorem (see Example 2.3.18) enables us to quantify how small θ must be for this approximation.

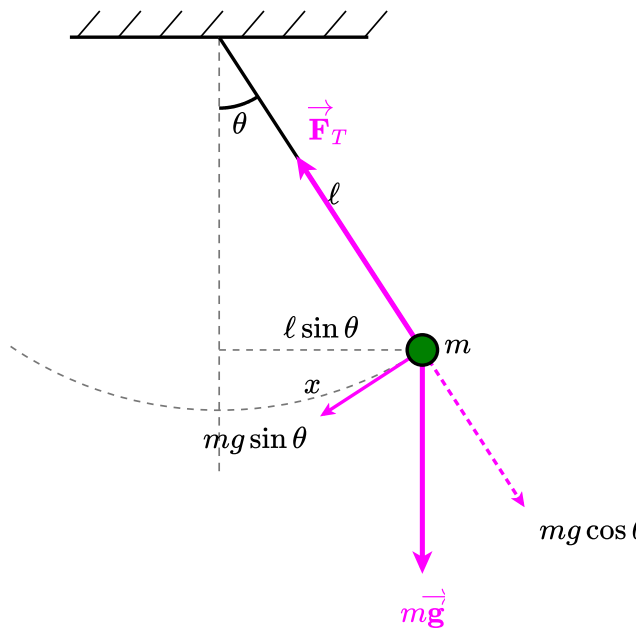


Figure 2.3.17 Forces on a simple pendulum. The weight $m\vec{g}$ resolves into a component $mg \cos \theta$ along the string (balanced by the tension $\vec{F}_T$) and a component $mg \sin \theta$ tangent to the arc, which acts as the restoring force.

An animation of these forces — the weight resolving into its radial and tangential components as the pendulum swings — accompanies the online version of these notes.

Example 2.3.18 Simple Harmonic Motion of a Pendulum as an Approximation. Use the remainder theorem to analyze the claim made in Giancoli's textbook that for small angles, $\sin \theta \approx \theta$. Specifically, show that the error is less than 1% for angles below 15° .

Here is the exact quote from Giancoli's textbook:

“For angles less than 15° , the difference between θ (in radians) and $\sin \theta$ is less than 1%.”

Solution.

workspace



2.3.4.4 *Approximating the value of π using Taylor series

Starred section. This one is for the interested reader. It will not be examined.

In this subsection, we will discuss how to approximate the value of π using Taylor series. We will use the Taylor series for $\arctan(x)$, which centered at $a = 0$ is

$$\begin{aligned}\arctan(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1} \\ &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots, \quad |x| \leq 1.\end{aligned}\tag{2.3.17}$$

Example 2.3.19 Approximating π using the Taylor series for $\arctan(x)$. Use the Taylor series for $\arctan(x)$ to approximate the value of π . Use Taylor's Remainder Theorem to find an upper bound for the error in this approximation, and show that the error decreases as the order n of the Taylor polynomial increases.

Solution.

workspace

For the record: the derivatives of $f(x) = \arctan(x)$ satisfy

$$\left| f^{(n+1)}(t) \right| \leq n! \quad \text{for all } t \in \mathbb{R}, \quad (2.3.18)$$

and the resulting error bound is

$$|\pi - 4T_n(1)| \leq \frac{4}{n+1}. \quad (2.3.19)$$

The bound goes to zero as $n \rightarrow \infty$, but very slowly: to guarantee an error of at most 0.01 one needs $n \geq 399$. $\blacktriangle$

As we saw in the previous example, Taylor series approached π very slowly. In the project below, we will see how to use the so-called Euler's formula to approximate π much faster. As you may know, there are many other methods to approximate π , which we will not cover here.

Project 2.3.4.1 Computing π with Euler's identity. In [Example 2.3.19](#) we approximated π by evaluating the Taylor series (2.3.17) at $x = 1$, and Taylor's Remainder Theorem gave the error bound (2.3.19), which decreases very slowly. In this guided problem we compute π far more efficiently using **Euler's identity**

$$\arctan\left(\frac{1}{2}\right) + \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}, \quad (2.3.20)$$

which lets us evaluate the Taylor series at the small arguments $x = \frac{1}{2}$ and $x = \frac{1}{3}$, where it converges much faster.

(a) *Proving Euler's identity.*

Let $\alpha = \arctan\left(\frac{1}{2}\right)$ and $\beta = \arctan\left(\frac{1}{3}\right)$. Use the addition formula

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

to prove that $\alpha + \beta = \frac{\pi}{4}$.

$\tan(\alpha + \beta) =$

why $\alpha + \beta$ must be $\pi/4$ and not another angle:

(b) *The approximation.*

Let $T_n(x)$ be the Taylor polynomial of order n for $\arctan(x)$ centered at 0. Use Euler's identity (2.3.20) to explain why

$$\pi \approx 4 \left[T_n\left(\frac{1}{2}\right) + T_n\left(\frac{1}{3}\right) \right].$$

Write out this approximation explicitly for $n = 3$.

why the approximation holds:

the case $n = 3$, as a number:

(c) *Bounding the error with the Remainder Theorem.*

In Example 2.3.19 we showed that the derivatives of $f(x) = \arctan(x)$ satisfy (2.3.18). Use (2.3.4) to show that for $|x| \leq 1$

$$|\arctan(x) - T_n(x)| \leq \frac{|x|^{n+1}}{n+1},$$

and conclude that

$$|\pi - 4 \left[T_n\left(\frac{1}{2}\right) + T_n\left(\frac{1}{3}\right) \right]| \leq \frac{4}{n+1} \left[\left(\frac{1}{2}\right)^{n+1} + \left(\frac{1}{3}\right)^{n+1} \right].$$

the bound on $|\arctan(x) - T_n(x)|$:

combining the two remainders:

(d) How much better is it?

Evaluate the error bound for $n = 9$ and compare it with the bound $\frac{4}{10} = 0.4$ obtained in [Example 2.3.19](#) for the same order. Then find the smallest n for which the bound guarantees an error of at most 10^{-6} .

the bound at $n = 9$, versus 0.4:

the smallest n with bound at most 10^{-6} :

An animation of the approximation $4[T_n(\frac{1}{2}) + T_n(\frac{1}{3})]$ converging to π , with its error compared against the much slower method at $x = 1$, accompanies the online version of these notes.

2.3.5 Proof of the remainder theorem

In this subsection, we will provide a proof of the remainder theorem.

Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a . We want to show that for each $x \in I$ and for each positive integer n , there exists a number c between a and x such that

$$f(x) = p_n(x) + R_n(x),$$

where

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k,$$

and

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

To prove this, we will first prove the following lemma:

Lemma 2.3.20 *Let $f(x)$ be a function that is $(n+1)$ -times differentiable. Also, suppose that $f(a) = f'(a) = f''(a) = \dots = f^{(n)}(a) = 0$ and $f(x) \neq 0$, where*

$x \neq a$. Then there exists a number c between a and x such that $f^{(n+1)}(c) = 0$.

Proof.

workspace

Remark 2.3.21 The error function and its derivatives are zero at the point of expansion. Note that for the error function $R_n(x) = f(x) - p_n(x)$, we have $R_n(a) = R'_n(a) = R''_n(a) = \dots = R_n^{(n)}(a) = 0$.

Completing the proof.

workspace

Chapter 3

3D Space and Vector Algebra

3.1 Three-Dimensional Coordinate Systems

Objectives

After completing this section, you should be able to:

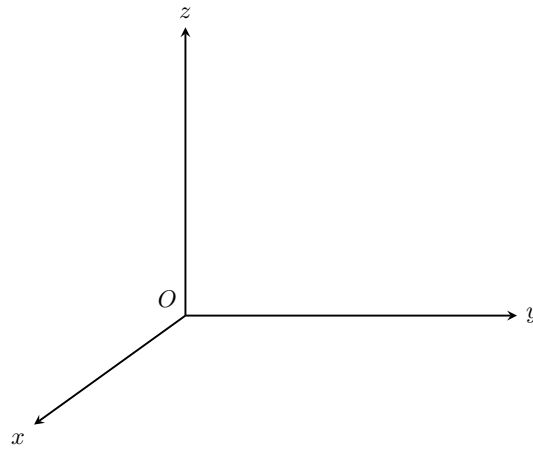
- Locate a point $P(a, b, c)$ in space using the rectangular (Cartesian) coordinate system, and check that the axes are arranged according to the right-hand rule.
- Identify the three coordinate planes and the eight octants, and recognize the surfaces described by equations such as $x = a$, $y = b$, and $z = c$.
- Compute the distance between two points in space, and write the equation of a sphere with a given center and radius.
- Describe the region of space determined by a collection of equations and inequalities, and use completing the square to recognize the equation of a sphere.

3.1.1 The Rectangular Coordinate System

To locate a point in the plane we need two numbers; to locate a point in space we need three. In this section we set up the rectangular coordinate system in space, which is the setting for everything that follows. We then measure distance, describe spheres, and practice translating between an equation and the surface or region it represents.

We will study the Cartesian coordinate system, which is also known as the **rectangular coordinate system**. We choose a point O in space, called the **origin**, and three mutually perpendicular lines through it, called the **coordinate axes** and labelled the x -, y -, and z -axis. A point P in space is then described by an ordered triple (a, b, c) , where a , b , and c are the signed distances from the origin along the three axes.

On the axes below, mark off a units along the x -axis, then b units parallel to the y -axis, then c units parallel to the z -axis, and label the point $P(a, b, c)$ you reach.



The three axes must follow the **right-hand rule**: if the index finger of the right hand points along the positive x -axis and the middle finger points along the positive y -axis, then the thumb points along the positive z -axis. [Figure 3.1.1](#) illustrates this convention; the square corner marks in that figure are a reminder that the three axes are mutually orthogonal, so each pair of axes meets at a right angle. Every coordinate system in this course is arranged this way.

The right-hand rule will return in the section on the cross product, where we use it in a second form to determine the direction of the cross product $\mathbf{u} \times \mathbf{v}$ of two vectors.

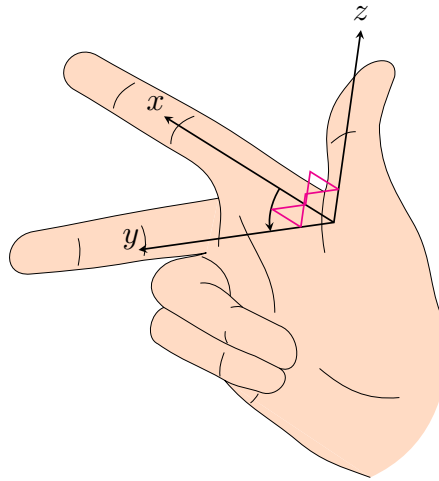
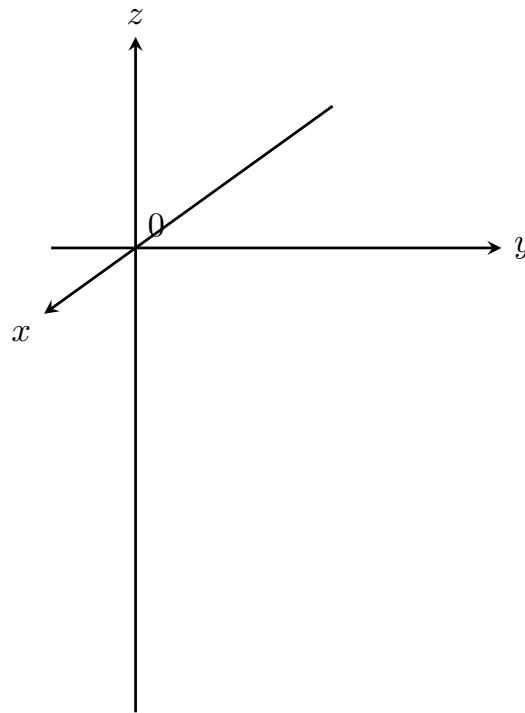


Figure 3.1.1 The right-hand rule. With the index finger of the right hand along the positive x -axis and the middle finger along the positive y -axis, the thumb points along the positive z -axis. The square corner marks record that the three axes are mutually orthogonal: each pair of axes meets at a right angle, even though the drawing on the page must distort those angles.

Now plot the point $(-4, 3, -5)$ on the axes below.



3.1.2 Coordinate Planes and Octants

The three coordinate axes determine three **coordinate planes**. The xy -plane is the plane containing the x - and y -axes, and it is described by the single equation $z = 0$; similarly the xz -plane is $y = 0$ and the yz -plane is $x = 0$. These three planes are shown in [Figure 3.1.2](#).

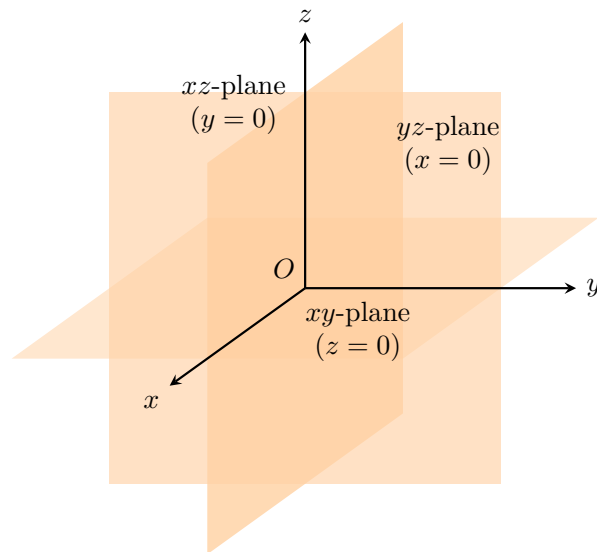


Figure 3.1.2 The three coordinate planes: the xy -plane ($z = 0$), the xz -plane ($y = 0$), and the yz -plane ($x = 0$). They divide space into eight octants.

In the plane the two axes create four quadrants; in space the three coordinate planes create eight **octants**. The **first octant** is the one in which x , y , and z are all positive.

From [Figure 3.1.2](#), record the sign pattern of the octant a point lies in.

First octant, signs of (x, y, z) :

The octant directly below it:

Octants with a negative coordinate:

More generally, equations such as $x = a$, $y = b$, and $z = c$ represent planes parallel to the yz -, xz -, and xy -planes respectively. Each of these equations places one restriction on a point of space and leaves the other two coordinates free, so each describes a plane. Consequently a point can be thought of as the intersection of three such planes: the point $P(1, 1, 1)$ of Figure 3.1.3 is the intersection of the planes $x = 1$, $y = 1$, and $z = 1$.

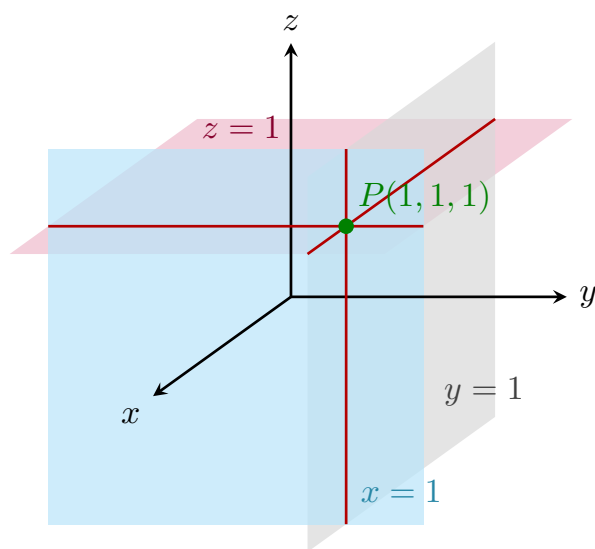


Figure 3.1.3 A point as the intersection of three planes. The planes $x = 1$, $y = 1$, and $z = 1$ meet pairwise in three lines, and all three lines meet in the single point $P(1, 1, 1)$.

3.1.3 Distance Between Two Points

The distance between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is denoted by $\|P_1P_2\|$ and can be computed with the Pythagorean theorem. As Figure 3.1.4 shows, the segment P_1P_2 is the diagonal of a rectangular box whose edges are parallel to the coordinate axes.

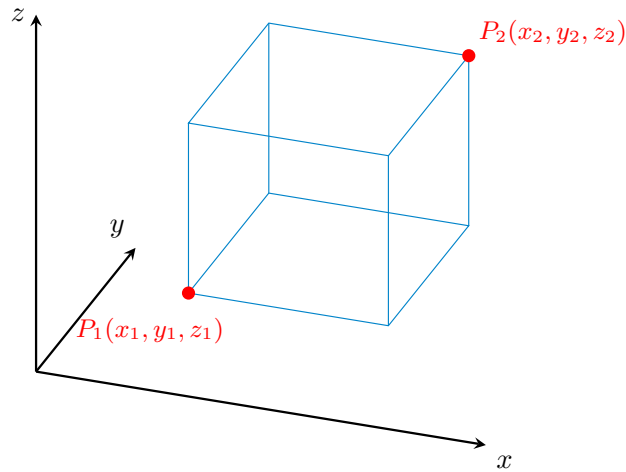


Figure 3.1.4 The segment P_1P_2 is the diagonal of a box with edges parallel to the coordinate axes.

Derive the distance formula from that box.

Step 1, in the base: $\|P_1B\|^2 =$

Step 2, up the vertical edge: $\|P_1P_2\|^2 =$

Substituting, $\|P_1P_2\| =$

Record the formula you obtained here, so it is easy to find later.

Distance in Space.

For $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$,

$$\|P_1P_2\| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}. \quad (3.1.1)$$

The video in [Figure 3.1.5](#) builds the box of [Figure 3.1.4](#) one edge at a time and applies the Pythagorean theorem twice, in the same two steps. Use it to check the derivation you just wrote.

BUG: PREVIEW NOT HANDLED



Standalone

Figure 3.1.5 Deriving the distance formula in space by applying the Pythagorean theorem twice: first in the base triangle P_1AB , then in the vertical triangle P_1BP_2 .

3.1.4 Spheres

A **sphere** of radius a is the set of all points $P(x, y, z)$ whose distance from a fixed center $P_0(x_0, y_0, z_0)$ equals a , as in [Figure 3.1.6](#).

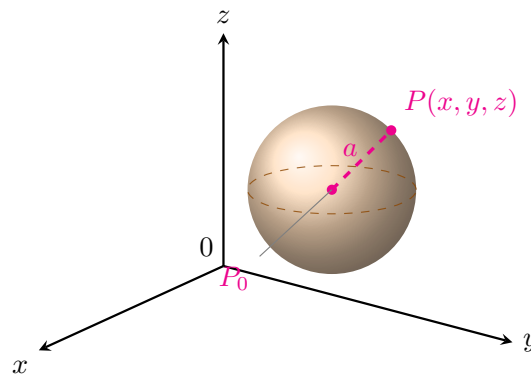


Figure 3.1.6 A sphere of radius a centered at $P_0 = P_0(x_0, y_0, z_0)$: the set of all points $P(x, y, z)$ at distance a from the center.

Turn that description into an equation.

The condition, before squaring:

Equation of the sphere:

Equation of a Sphere.

The sphere of radius a centered at $P_0(x_0, y_0, z_0)$ is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2. \quad (3.1.2)$$

Example 3.1.7 Recognizing a Sphere by Completing the Square.

Describe the geometric surface represented by the following equation.

$$x^2 + y^2 + z^2 + 2x + 4y = 0$$

Solution.

Completing the square:

Center:

Radius:



3.1.5 Describing Regions in Space

A single equation in x , y , and z normally describes a surface, two simultaneous equations describe the curve where two surfaces meet, and inequalities describe solid regions. The next example collects several of these situations.

Example 3.1.8 Describing Equations and Inequalities Geometrically.

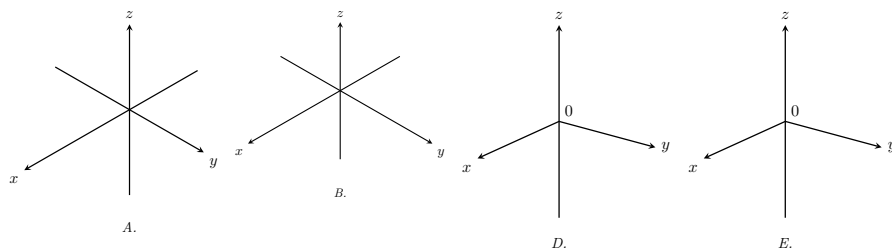
Describe the geometrical meaning of the following equalities and inequalities.

- A. $x = 1$
- B. $x = 1, y = 2$
- C. $x^2 + y^2 + z^2 = 1$
- D. $1 \leq x^2 + y^2 + z^2 \leq 4, z \leq 0$
- E. $x^2 + y^2 = 1$
- F. $x^2 + y^2 = 1, -1 \leq z \leq 1$
- G. $x^2 + y^2 = 1, z = 3$

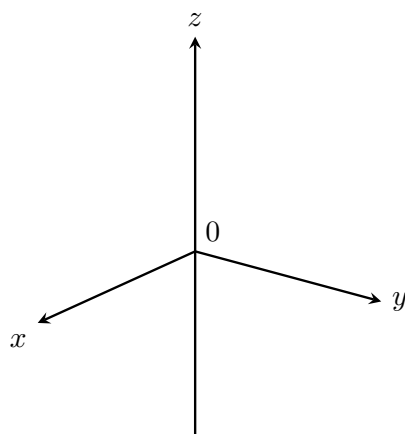
Solution.

A.
B.
C.
D.
E.
F.
G.

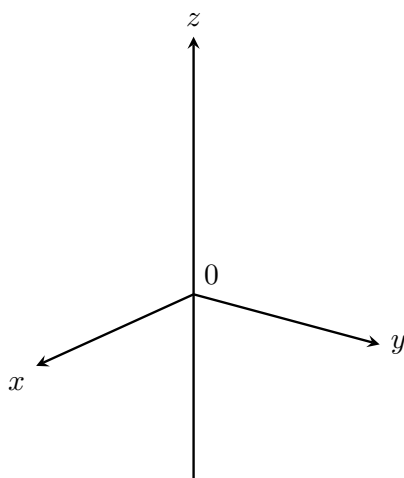
Now sketch parts **A**, **B**, **D**, and **E** on the four sets of axes below.



Parts **F** and **G** are both built from the cylinder of part **E**. Sketch them on the axes below, and say in one line what changed.



$$F. x^2 + y^2 = 1, -1 \leq z \leq 1$$



$$G. x^2 + y^2 = 1, z = 3$$



3.2 Vectors

Objectives

After completing this section, you should be able to:

- Explain the difference between a scalar quantity and a vector quantity, and describe a vector by its initial and terminal points.
- Write a vector in component form, find the standard position vector of $\overrightarrow{PQ}$, and compute the length $\|\mathbf{v}\|$ of a vector.
- Add vectors and multiply them by scalars, both algebraically and geometrically using the parallelogram law.
- Write a vector as a linear combination of the standard unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$, and find the unit vector in the direction of a given vector.

3.2.1 Definitions, Terminology, and Notation

Some quantities can be described using real numbers alone, such as time or temperature. To describe other quantities we need a direction as well as a magnitude. Consider the velocity of a car: we need to know both the speed of the car and its direction of motion to specify its velocity completely. For instance, we might say that the car is moving with speed 65 miles per hour in the direction of north. Such quantities are referred to as **vector quantities**, and they are the subject of this section.

A **vector** is a directed line segment from an **initial point** A to a **terminal point** B , and is denoted by $\overrightarrow{AB}$. We denote the length of a vector by $\|\overrightarrow{AB}\|$. See [Figure 3.2.1](#).

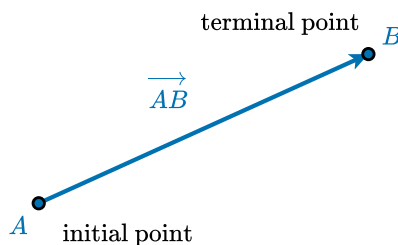


Figure 3.2.1 A vector is a directed line segment. It carries two pieces of information: a direction, shown by the arrowhead, and a magnitude, which is its length $\|\vec{AB}\|$.

Two vectors are **equal** if they have the same direction and the same magnitude. Notice that this definition says nothing about where a vector is located: a vector may be moved around the plane freely, and as long as its direction and length are unchanged it is the same vector. The vectors shown in Figure 3.2.2 are all equal to each other, $\vec{AB} = \vec{CD} = \vec{OP} = \vec{EF}$.

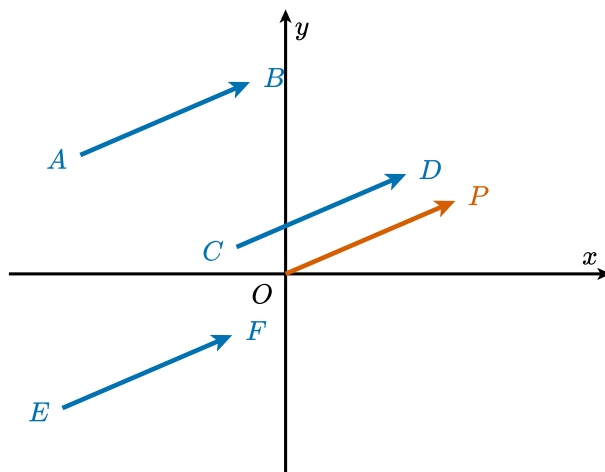


Figure 3.2.2 Four equal vectors. They have different initial points, but the same direction and the same length, so $\vec{AB} = \vec{CD} = \vec{OP} = \vec{EF}$. Only $\vec{OP}$ begins at the origin.

Only one of the vectors in Figure 3.2.2 starts from the origin. We say that $\vec{OP}$ is in **standard position**, and we denote it by a bold letter such as $\mathbf{v} = \vec{OP}$, which is also referred to as the **standard position vector**.

In 2D, if the initial point of a vector is the origin, i.e. $(0, 0)$, and its final point is (v_1, v_2) , then the **component form** of the vector is $\mathbf{v} = \langle v_1, v_2 \rangle$. In 3D, if the initial point of a vector is the origin, i.e. $(0, 0, 0)$, and its final point is (v_1, v_2, v_3) , then the component form of the vector is $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$.

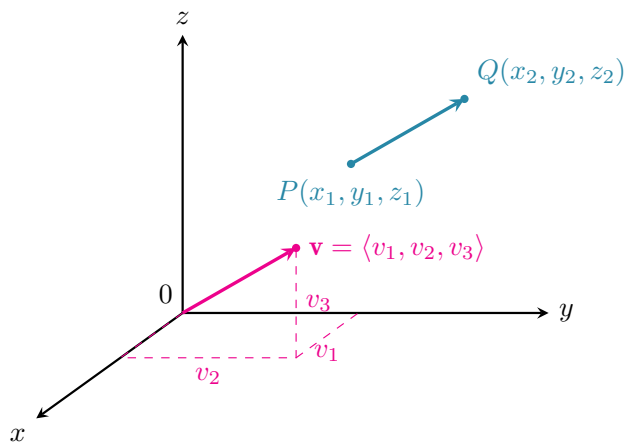


Figure 3.2.3 The vector $\overrightarrow{PQ}$ and its standard position vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. The two arrows are equal as vectors; only the second one begins at the origin.

Read the components of $\mathbf{v} = \overrightarrow{PQ}$ off [Figure 3.2.3](#).

Components: $\mathbf{v} = \langle v_1, v_2, v_3 \rangle =$

Length: $\|\mathbf{v}\| =$

Components and Length.

For $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, the standard position vector of $\overrightarrow{PQ}$ is

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle, \quad (3.2.1)$$

and its magnitude is

$$\begin{aligned} \|\mathbf{v}\| &= \|\overrightarrow{PQ}\| = \sqrt{v_1^2 + v_2^2 + v_3^2} \\ &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}. \end{aligned} \quad (3.2.2)$$

Example 3.2.4 Length and Standard Position Vector.

- Compute the length of the vector with the initial point $P(-1, 0, 1)$ and the final point $Q(1, 1, -1)$.
- Find the standard position vector corresponding to this vector.

Solution.

A. $\|\overrightarrow{PQ}\| =$

B. $\mathbf{v} =$

Check: $\|\mathbf{v}\| =$



3.2.2 Vector Algebra Operations

Definition 3.2.5 Vector Addition. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. The **sum** of $\mathbf{u}$ and $\mathbf{v}$ is the vector

$$\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle. \quad (3.2.3)$$



Parallelogram law. The parallelogram law tells us how to add two vectors geometrically, and is described in [Figure 3.2.6](#). Placing the tail of $\mathbf{v}$ at the head of $\mathbf{u}$ produces the sum as the third side of a triangle; equivalently, drawing both vectors from a common initial point makes the sum the diagonal of the parallelogram they span.

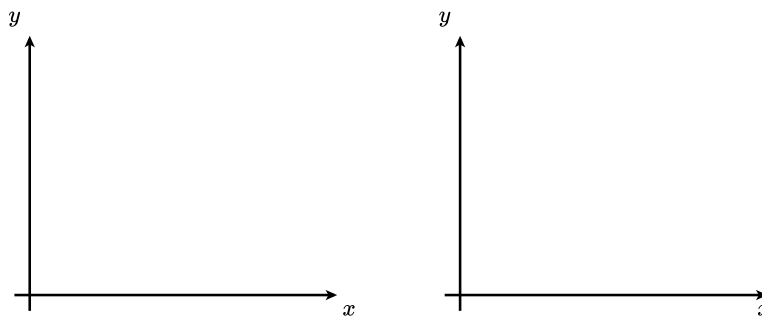


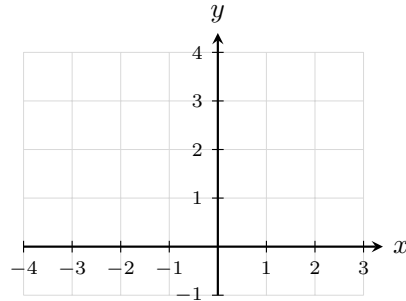
Figure 3.2.6 The parallelogram law: head to tail on the left, the diagonal of the parallelogram on the right.

Example 3.2.7 Adding Two Vectors. Add the following two vectors geometrically and algebraically.

$$\mathbf{u} = \langle 1, 1 \rangle \quad \text{and} \quad \mathbf{v} = \langle -2, 1 \rangle$$

Solution.

$$\mathbf{u} + \mathbf{v} =$$



Definition 3.2.8 Scalar Multiplication. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and let $k \in \mathbb{R}$ be a scalar. The **scalar multiple** $k\mathbf{u}$ is the vector

$$k\mathbf{u} = \langle ku_1, ku_2, ku_3 \rangle. \quad (3.2.4)$$

Note that both definitions can be reduced to the 2D case simply by removing the last components of the vectors.

Scaling a vector by k scales its length by $|k|$. Show this.

$$\|k\mathbf{u}\| =$$

Length of a Scalar Multiple.

$$\|k\mathbf{u}\| = |k| \|\mathbf{u}\|. \quad (3.2.5)$$

The vector $-\mathbf{u}$ has the same length as $\mathbf{u}$; however, the two vectors point in opposite directions. In general the length of $k\mathbf{u}$ is $|k|$ times the length of $\mathbf{u}$, and if $k > 0$ the two vectors point in the same direction, whereas if $k < 0$ they point in opposite directions. [Figure 3.2.9](#) shows several examples.

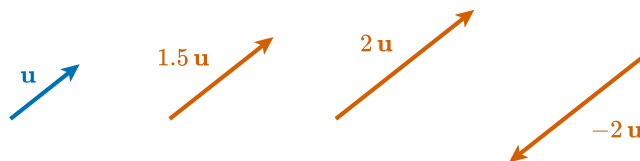


Figure 3.2.9 Scalar multiples of $\mathbf{u}$. Multiplying by 1.5 or 2 stretches the vector without turning it; multiplying by -2 stretches it and reverses its direction.

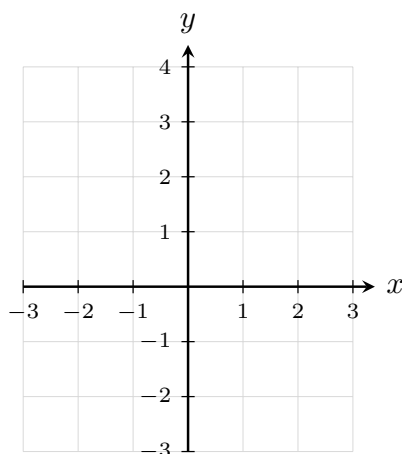
Definition 3.2.10 Vector Subtraction. Subtraction is handled with the same idea: the **difference** $\mathbf{u} - \mathbf{v}$ means

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}), \quad (3.2.6)$$

so we reverse $\mathbf{v}$ and then add. Geometrically, $\mathbf{u} - \mathbf{v}$ is the vector that points from the head of $\mathbf{v}$ to the head of $\mathbf{u}$ when both are drawn from a common initial point. $\blacklozenge$

Example 3.2.11 Adding and Subtracting Geometrically. Given the two vectors $\mathbf{u} = \langle 1, 1 \rangle$ and $\mathbf{v} = \langle -1, 2 \rangle$, find $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ geometrically.

Solution.



$\mathbf{u} + \mathbf{v} =$

$\mathbf{u} - \mathbf{v} =$

$\blacktriangle$

The operations satisfy the familiar algebraic rules collected in [Theorem 3.2.12](#).

Theorem 3.2.12 Properties of Vector Operations. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors and a, b be scalars. Then

- | | |
|--|--|
| 1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ | 2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ |
| 3. $\mathbf{u} + \mathbf{0} = \mathbf{u}$ | 4. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ |
| 5. $0\mathbf{u} = \mathbf{0}$ | 6. $1\mathbf{u} = \mathbf{u}$ |
| 7. $a(b\mathbf{u}) = (ab)\mathbf{u}$ | 8. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ |
| 9. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$ | |

3.2.3 Unit Vectors

Vectors of length one are referred to as **unit vectors**. The **standard unit vectors** are the ones pointing in the positive direction of x , y , and z , and they are denoted by $\hat{i}$, $\hat{j}$, and $\hat{k}$ respectively, as in Figure 3.2.13.

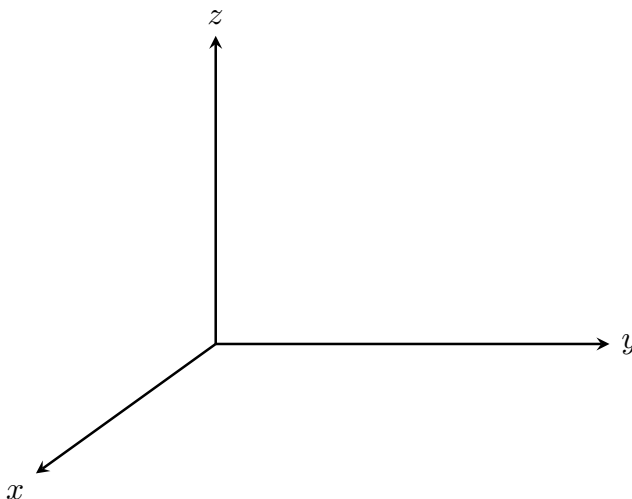


Figure 3.2.13 Empty axes for the standard unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$, and for the decomposition $\mathbf{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ of a vector into components along the axes.

Combining (3.2.3) and (3.2.4), any vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ can be written as a linear combination of $\hat{i}$, $\hat{j}$, and $\hat{k}$ as

$$\mathbf{a} = \langle a_1, a_2, a_3 \rangle = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}. \quad (3.2.7)$$

To find a unit vector pointing in the same direction as a nonzero vector $\mathbf{a}$, we divide $\mathbf{a}$ by its own length. We write $\hat{a}$, read “a hat”, for the resulting unit vector:

$$\hat{a} = \frac{\mathbf{a}}{\|\mathbf{a}\|}. \quad (3.2.8)$$

From now on a hat always marks a vector of length one, as it does for the standard unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$.

Check that (3.2.8) really does produce a unit vector.

$$\|\hat{a}\| =$$

Example 3.2.14 Standard Unit Vectors and a Unit Direction. Consider the two vectors $\mathbf{v} = \langle 1, 0, 1 \rangle = \hat{i} + \hat{k}$ and $\mathbf{u} = \langle -1, 1, 0 \rangle = -\hat{i} + \hat{j}$.

- A. Express $\mathbf{u} + \mathbf{v}$ in terms of the standard unit vectors.
- B. Find the unit vector in the direction of $\mathbf{u} + \mathbf{v}$.

Solution.

$$A. \mathbf{u} + \mathbf{v} =$$

$$\|\mathbf{u} + \mathbf{v}\| =$$

$$B. \widehat{\mathbf{u} + \mathbf{v}} =$$



3.3 The Dot Product

Objectives

After completing this section, you should be able to:

- Understand the dot product as a way of combining two vectors into a single number that measures the extent to which they point in the same direction.
- Interpret the sign of the dot product geometrically—positive for an acute angle, negative for an obtuse angle, and zero for perpendicular vectors—and recognize orthogonality as the vanishing of the dot product.
- Understand how the dot product encodes the angle between two vectors, connecting the algebra of vectors to the Law of Cosines.
- Interpret the projection of one vector onto another as the component of one vector in the direction of the other, and understand the decomposition of a vector into parts parallel and perpendicular to a given direction.
- Understand work as a physical measure of the effect of a force along a direction of motion, and appreciate why the dot product is the natural tool for expressing it.
- Explain how the dot product is used in data science to measure similarity: represent preferences as vectors, compute the cosine similarity and the angular distance between them, interpret the result, and extend the computation from two dimensions to $\mathbb{R}^n$.

3.3.1 The Dot Product and the Angle between Two Vectors

Consider two vectors $\mathbf{u}$ and $\mathbf{v}$, and let $\mathbf{w} = \mathbf{u} - \mathbf{v}$, as shown in Figure 3.3.1. By the law of cosines,

$$\|\mathbf{w}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta. \quad (3.3.1)$$

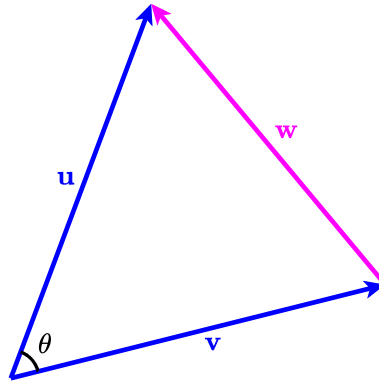


Figure 3.3.1 The vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w} = \mathbf{u} - \mathbf{v}$, together with the angle θ between $\mathbf{u}$ and $\mathbf{v}$.

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[Standalone](#)

Figure 3.3.2 Proving the law of cosines: drop a perpendicular from the tip of $\mathbf{u}$ and apply the Pythagorean theorem to the right triangle whose hypotenuse is $\mathbf{w} = \mathbf{u} - \mathbf{v}$.

Now compute $\|\mathbf{w}\|^2$ a second way, in components, and compare the two answers.

In components: $\|\mathbf{w}\|^2 =$

Setting the two expressions equal:

Solving for $\cos \theta$:

Theorem 3.3.3 Angle between Two Vectors. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ be two nonzero vectors, and let θ , with $0 \leq \theta \leq \pi$, be the angle between them. Then

$$\begin{aligned} \cos \theta &= \frac{u_1v_1 + u_2v_2 + u_3v_3}{\|\mathbf{u}\|\|\mathbf{v}\|} \\ \iff \theta &= \cos^{-1} \left(\frac{u_1v_1 + u_2v_2 + u_3v_3}{\|\mathbf{u}\|\|\mathbf{v}\|} \right). \end{aligned} \quad (3.3.2)$$

Definition 3.3.4 Dot Product. The term $u_1v_1 + u_2v_2 + u_3v_3$ in the numerator of (3.3.2) is known as the **dot product** of the two vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$, and is denoted $\mathbf{u} \cdot \mathbf{v}$.

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3. \quad (3.3.3)$$



Example 3.3.5 Computing a dot product. Let $\mathbf{u} = \langle 1, 2, 3 \rangle$ and $\mathbf{v} = \langle -2, 0, 4 \rangle$. Compute $\mathbf{u} \cdot \mathbf{v}$, and use it to find the angle between $\mathbf{u}$ and $\mathbf{v}$.

Solution.

$\mathbf{u} \cdot \mathbf{v} =$	<i>so the angle is:</i>
$\ \mathbf{u}\ =$	$\ \mathbf{v}\ =$
$\cos \theta =$	
$\theta =$	

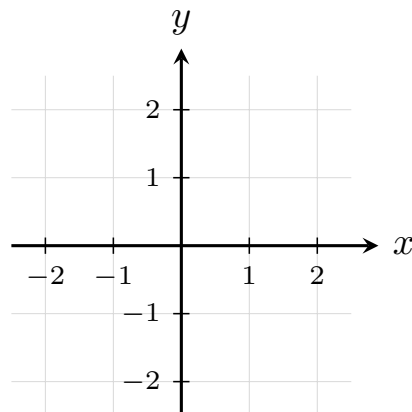


Remark 3.3.6 Orthogonal vectors. If the two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, then $\theta = \pi/2$, which means $\mathbf{u} \cdot \mathbf{v} = 0$. Conversely, if $\mathbf{u}$ and $\mathbf{v}$ are two vectors such that $\mathbf{u} \cdot \mathbf{v} = 0$, then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

$$\mathbf{u} \cdot \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v}. \quad (3.3.4)$$

Example 3.3.7 A right triangle detected with the dot product. Consider the triangle with vertices $A(1, -1)$, $B(1, 1)$, and $C(2, 0)$. Show that this is a right triangle using the dot product and the orthogonality criterion (3.3.4).

Solution.



<i>The two vectors and their dot product:</i>
<i>Conclusion:</i>



Example 3.3.8 The remaining angles of the triangle. For the triangle of Example 3.3.7, calculate the remaining angles.

Solution.

The angle at A:

The angle at B:



Properties of the Dot Product.

Fact 3.3.9 If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are any vectors and c is a scalar, then

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
4. $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$
5. $\mathbf{0} \cdot \mathbf{u} = 0$

Prove properties 1, 3, and 4.

1.

3.

4.

3.3.2 Projection of $\mathbf{u}$ in the Direction of $\mathbf{v}$

The dot product measures how much two vectors point in the same direction. This section turns that measurement into a vector: given $\mathbf{u}$ and $\mathbf{v}$, we ask how much of $\mathbf{u}$ lies along $\mathbf{v}$. The answer is the **projection** of $\mathbf{u}$ onto $\mathbf{v}$, and it lets us split any vector into a part that points along $\mathbf{v}$ and a part perpendicular to it. That decomposition is what makes the dot product useful in practice—it is how we find the component of a force along a direction of motion, as in [Subsection 3.3.3](#), or the component of gravity down a slope.

The projection of $\mathbf{u}$ in the direction of $\mathbf{v}$ is denoted $\text{proj}_{\mathbf{v}}\mathbf{u}$, and is shown in [Figure 3.3.10](#).

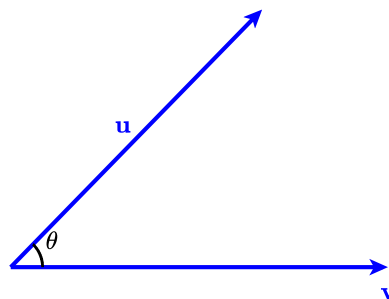


Figure 3.3.10 The vectors $\mathbf{u}$ and $\mathbf{v}$, for the projection $\text{proj}_{\mathbf{v}}\mathbf{u}$ of $\mathbf{u}$ in the direction of $\mathbf{v}$ and the perpendicular component $\mathbf{u} - \text{proj}_{\mathbf{v}}\mathbf{u}$.

Build the projection formula from that picture.

Length $\times$ direction:

Replacing $\cos \theta$:

$\text{proj}_{\mathbf{v}} \mathbf{u} =$

Fact 3.3.11 Projection Formula.

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \right) \mathbf{v}. \quad (3.3.5)$$

Writing $\mathbf{u}$ as Two Vectors, One Parallel and One Perpendicular to $\mathbf{v}$. As can be seen in [Figure 3.3.10](#), $\text{proj}_{\mathbf{v}} \mathbf{u}$ is parallel to $\mathbf{v}$, and the vector $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$ is perpendicular to it. Also, the sum of these two vectors equals $\mathbf{u}$. This means we can write $\mathbf{u}$ as

$$\mathbf{u} = \underbrace{\hspace{2cm}}_{\text{parallel to } \mathbf{v}} + \underbrace{\hspace{2cm}}_{\text{orthogonal to } \mathbf{v}}. \quad (3.3.6)$$

$\mathbf{u} = \text{proj}_{\mathbf{v}} \mathbf{u} + (\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}) =$

with the projection formula:

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Figure 3.3.12 The projection of $\vec{u}$ onto $\vec{v}$ as a shadow cast by light from above.

Example 3.3.13 Decomposing a vector into parallel and perpendicular parts. Consider the two vectors $\mathbf{v} = \hat{i} + \hat{j}$ and $\mathbf{u} = 2\hat{i} + 2\hat{j} + 2\hat{k}$. Write $\mathbf{u}$ as a sum of two vectors, one of which is parallel to $\mathbf{v}$ and the other perpendicular to it.

Solution.

$\mathbf{u} \cdot \mathbf{v} =$	$\ \mathbf{v}\ =$
$\mathbf{u}_{\parallel} =$	
$\mathbf{u}_{\perp} =$	
<i>Check:</i> $\mathbf{u}_{\perp} \cdot \mathbf{v} =$	

To achieve a visual understanding of this decomposition, watch the video below.

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Figure 3.3.14 Decomposing $\mathbf{u} = 2\hat{i} + 2\hat{j} + 2\hat{k}$ into $\mathbf{u}_{\parallel} = 2\hat{i} + 2\hat{j}$, lying in the xy -plane along $\mathbf{v}$, and $\mathbf{u}_{\perp} = 2\hat{k}$, pointing straight up.



Checkpoint 3.3.15 Concept Check. Everything in (3.3.6) rests on the claim read off from Figure 3.3.10: that $\text{proj}_{\mathbf{v}}\mathbf{u}$ is parallel to $\mathbf{v}$, and that $\mathbf{u} - \text{proj}_{\mathbf{v}}\mathbf{u}$ is perpendicular to $\mathbf{v}$. A picture is not a proof. Explain why each half of the claim is true.

parallel half:

perpendicular half:

3.3.3 Application in Physics: Work

Consider the scenario in which a constant force $\mathbf{F}$ causes an object to move from point P to point Q , as shown in Figure 3.3.18. The vector $\overrightarrow{PQ}$ is often denoted $\mathbf{D}$ and is referred to as the **displacement vector**. The work done by the force is then

Definition 3.3.16 Work.

$$\text{Work} = \mathbf{F} \cdot \mathbf{D} = \|\mathbf{D}\| \|\mathbf{F}\| \cos \theta, \quad (3.3.7)$$

where $\|\text{proj}_{\mathbf{D}}\mathbf{F}\| = \|\mathbf{F}\| \cos \theta$ is the magnitude of the force in the direction of motion, as computed in Subsection 3.3.2. $\blacklozenge$

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Figure 3.3.17 A constant force $\mathbf{F}$ acting on an object causing a displacement from A to B .

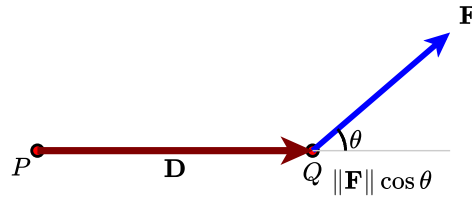


Figure 3.3.18 A force $\mathbf{F}$ displaces an object from P to Q along the displacement vector $\mathbf{D}$. The reference segment shows the magnitude of the force in the direction of motion, $\|\mathbf{F}\| \cos \theta$.

Example 3.3.19 Computing work done by a force. A force is given by the vector $\mathbf{F} = 8\hat{i} - 6\hat{j} + 9\hat{k}$ and moves a particle from the point $A(0, 10, 8)$ to the point $B(6, 12, 20)$. Find the work done.

Solution.

$\mathbf{D} =$

Work =

For more help, watch the video below.

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Figure 3.3.20 Set up the work computation: build the displacement vector first, then take the dot product. What do you get?



Example 3.3.21 Gravity on an incline. Suppose a mass of 10 kg is resting on an inclined plane. Gravity exerts a force equivalent to mg on the object, where $g = 9.8 \text{ m/s}^2$ is the gravitational acceleration. Suppose the incline is tilted at a 30° angle. Compute the components of the force that are parallel and perpendicular to the inclined plane.

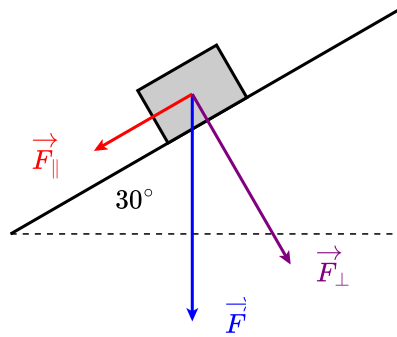


Figure 3.3.22 Forces acting on a mass on a 30° incline: the weight $\vec{F}$ resolves into a component $\vec{F}_\parallel$ down the slope and a component $\vec{F}_\perp$ into the surface, with $\vec{F} = \vec{F}_\parallel + \vec{F}_\perp$.

Solution.

$\ \vec{F}\ =$	$\vec{F} =$
$\hat{v} =$	
$\vec{F}_{\parallel} =$	
$\vec{F}_{\perp} =$	

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Figure 3.3.23 Decomposing gravity on a 30° incline: the parallel component $\vec{F}_{\parallel} = \text{proj}_{\hat{v}} \vec{F}$ pulls the mass down the slope, and the perpendicular component $\vec{F}_{\perp} = \vec{F} - \vec{F}_{\parallel}$ presses it into the surface.



3.3.4 Application in Data Science: Measuring Similarity in Tastes

Every time a streaming service tells you “because you watched *Sinners...*”, some piece of software has decided that your taste resembles the taste of other people in its database. Deciding whose taste resembles whose can be framed as a geometry problem, and one tool that can help us decide is the dot product.

The first step is to turn taste into a vector. Fix a list of movies and ask each person to rate every one of them on a scale from -5 to 5 , where 5 means “loved it”, -5 means “I want those two hours back”, and 0 means indifference. Reading the ratings off in the same order for everybody turns each person into a vector, and the question “do these two people have similar taste?” becomes

a question about the angle between two vectors, which by [Theorem 3.3.3](#) is exactly what the dot product measures.

Definition 3.3.24 Angular Distance and Cosine Similarity. Let $\mathbf{u}$ and $\mathbf{v}$ be two nonzero vectors. Their **angular distance** is the angle θ between them, obtained from [\(3.3.2\)](#),

$$d_A(\mathbf{u}, \mathbf{v}) = \theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \right). \quad (3.3.8)$$

The quantity $\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \cos \theta$ appearing inside is called the **cosine similarity** of $\mathbf{u}$ and $\mathbf{v}$. The smaller the angular distance, the more similar the two vectors. ♦

Angular distance asks “how nearly do these two vectors point the same way?” Notice that it is completely blind to the lengths of $\mathbf{u}$ and $\mathbf{v}$: multiplying either vector by a positive number leaves θ untouched. We will come back to what that means for moviegoers.

Two Movies at a Time. Two movies give vectors in $\mathbb{R}^2$, which we can draw. Take two of the ten films nominated for Best Picture at the 2026 Academy Awards, *Sinners* and *Hamnet*, and record each person’s rating of *Sinners* as the first component and of *Hamnet* as the second. Mehdi, Norm, and Popcorn rate them as follows:

$$\text{Mehdi: } \mathbf{u} = \langle 4, 5 \rangle$$

$$\text{Norm: } \mathbf{v} = \langle 5, 3 \rangle$$

$$\text{Popcorn: } \mathbf{w} = \langle -4, -2 \rangle.$$

Mehdi and Norm both enjoyed the two films; Popcorn sat through both of them and regretted it. [Figure 3.3.25](#) shows the three taste vectors.

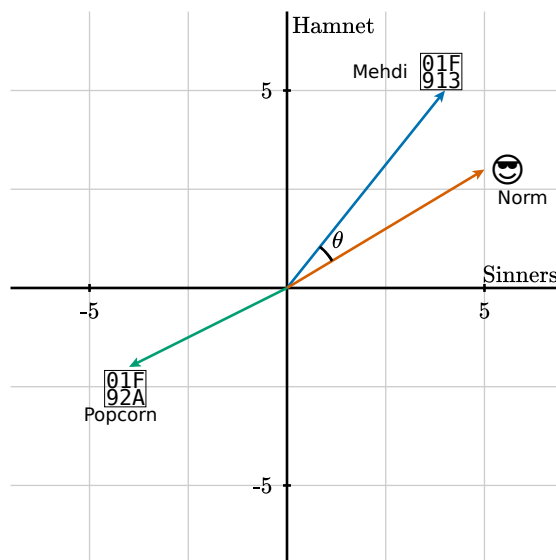


Figure 3.3.25 The taste vectors $\mathbf{u}$ (Mehdi), $\mathbf{v}$ (Norm), and $\mathbf{w}$ (Popcorn). The angle θ is the angular distance between $\mathbf{u}$ and $\mathbf{v}$.

Example 3.3.26 Who should watch a movie together? Using the taste vectors above, rank the three pairs by angular distance.

Solution.

Table 3.3.27 Angular distance between the three taste vectors.

Pair	$\mathbf{u} \cdot \mathbf{v}$	$\cos \theta$	d_A
Mehdi, Norm			
Mehdi, Popcorn			
Norm, Popcorn			

Lengths and dot products:

Who are the movie buddies, and what did the signs already tell you?



Angular Distance Ignores Enthusiasm. Suppose Ray, Vera, and Gus rate the same two films

Ray: $\mathbf{a} = \langle 2, 2 \rangle$

Vera: $\mathbf{b} = \langle 5, 4 \rangle$

Gus: $\mathbf{c} = \langle 5, 5 \rangle$,

as in [Figure 3.3.28](#).

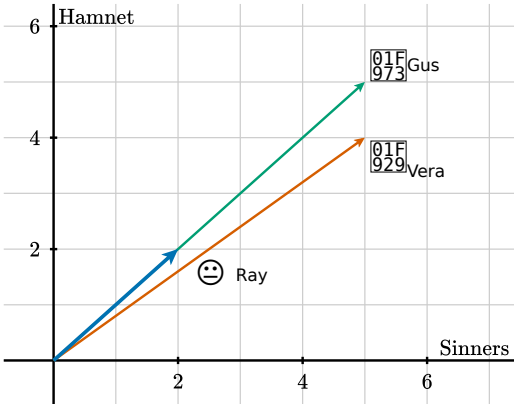


Figure 3.3.28 Ray, Vera, and Gus. Ray’s vector $\mathbf{a}$ lies along Gus’s vector $\mathbf{c}$, so their angular distance is 0, even though Gus handed out far more generous ratings.

Find $d_A(\mathbf{a}, \mathbf{c})$ and $d_A(\mathbf{b}, \mathbf{c})$.

$d_A(\mathbf{a}, \mathbf{c}) =$ *because*

$d_A(\mathbf{b}, \mathbf{c}) =$

What this says about enthusiasm:

More Movies, More Dimensions. Nothing above depended on there being exactly two movies. With n movies, a person's taste vector is $\mathbf{u} = \langle u_1, u_2, \dots, u_n \rangle$ in $\mathbb{R}^n$, the dot product of (3.3.3) becomes a sum of n products, $\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i$, and the length of a vector becomes $\|\mathbf{u}\| = \sqrt{\sum_{i=1}^n u_i^2}$. The angular distance of Definition 3.3.24 carries over verbatim:

$$d_A(\mathbf{u}, \mathbf{v}) = \cos^{-1} \left(\frac{\sum_{i=1}^n u_i v_i}{\sqrt{\sum_{i=1}^n u_i^2} \sqrt{\sum_{i=1}^n v_i^2}} \right). \quad (3.3.9)$$

We can no longer draw the picture, but every formula still applies, and the language of Remark 3.3.6 still means something: two taste vectors with $\mathbf{u} \cdot \mathbf{v} = 0$ are orthogonal, an angular distance of 90° , which here says that knowing one person's ratings tells you nothing about the other's. This is what makes the dot product useful in practice. A streaming service with 50,000 titles works in $\mathbb{R}^{50,000}$, and finding the users nearest to you is still nothing more than the arithmetic in (3.3.9).

Example 3.3.29 All ten nominees. List the ten 2026 Best Picture nominees alphabetically—*Bugonia*, *F1*, *Frankenstein*, *Hamnet*, *Marty Supreme*, *One Battle After Another*, *The Secret Agent*, *Sentimental Value*, *Sinners*, *Train Dreams*—and suppose Mehdi, Norm, and Popcorn rate all ten:

$$\begin{aligned} \mathbf{u} &= \langle 3, -1, 2, 5, 0, 5, 4, 3, 4, 2 \rangle \\ \mathbf{v} &= \langle 1, 4, 3, 3, 5, 4, 0, 1, 5, -2 \rangle \\ \mathbf{w} &= \langle -2, 3, -3, -2, 4, -1, -4, -5, -4, -3 \rangle. \end{aligned}$$

(The fourth and ninth entries are the *Hamnet* and *Sinners* ratings from before.) Which pair has the most similar taste?

Solution.

Table 3.3.30 Angular distance across all ten nominees.

Pair	$\mathbf{u} \cdot \mathbf{v}$	$\cos \theta$	d_A
Mehdi, Norm			
Mehdi, Popcorn			
Norm, Popcorn			

Lengths and dot products:

Compared with the two-movie table:

▲
Ten components are a lot to handle by hand, and a real recommender system deals with thousands. The cell below carries out the computation of [Example 3.3.29](#). Press *Evaluate* to run it and check the table you just filled in, then edit the ratings and run it again to see whose taste your own vector is closest to.

```
# ratings of the ten nominees, in alphabetical order
u = vector([ 3, -1,  2,  5,  0,  5,  4,  3,  4,  2])
v = vector([ 1,  4,  3,  3,  5,  4,  0,  1,  5, -2])
w = vector([-2,  3, -3, -2,  4, -1, -4, -5, -4, -3])

def cosine_similarity(a, b):
    return a.dot_product(b) / (a.norm() * b.norm())

# the angular distance, converted from radians to degrees
def angular_distance(a, b):
    return arccos(cosine_similarity(a, b)) * 180 / pi

for name, a, b in [("Mehdi, Norm", u, v),
                  ("Mehdi, Popcorn", u, w),
                  ("Norm, Popcorn", v, w)]:
    print(name, N(cosine_similarity(a, b), digits=3),
          N(angular_distance(a, b), digits=4))
```

```
Mehdi, Norm 0.549 56.71
Mehdi, Popcorn -0.762 139.6
Norm, Popcorn -0.0744 94.27
```

Chapter 4

Assignments and Review Problems

The ten assignments, the four review problem sets, and a sample past exam, in one place. Worked solutions appear beneath each problem once the assignment's due date has passed; solutions for the review sets are posted the same way.

Using the AI Tutor.

Stuck? A Socratic AI tutor is available on every assignment page. It knows the problems on the page and will coach you with questions and gentle hints — but it will never give away an answer. Here is how to use it.

Opening the tutor. On an assignment page, click the round “Tutor” button in the lower-right corner. A chat panel opens and stays in place as you scroll, so you can read a problem and talk to the tutor at the same time. The dash button in the panel's title bar minimizes it back to the button.

The tutor is available on each of the ten assignment pages. The review problem sets do not have it: those sets are practice for the exams, where you will be entirely on your own.

One-time setup. The first time you open the tutor it asks for a free Google AI Studio key (this takes about a minute):

1. Go to aistudio.google.com/apikey and sign in with a *personal* Google account — a regular Gmail. Do *not* use your @scu.edu account: the university has this service turned off, and you will see “Your account is managed by an organization.”
2. Click *Create API key* and copy it.
3. Paste it into the tutor.

The key is stored only in your own browser and is sent only to Google; this site never sees it. Treat the key as yours alone: anyone who has it can spend your free quota, so do not share it or paste it anywhere else.

Privacy: what to keep out of the chat. Everything you type into the tutor is sent to Google and handled under Google's terms, not the university's. On the free tier your conversations *may be read by*

people at Google and used to improve their products, and they are not covered by the protections your school records have. Nothing you type is private, so treat the panel as a public place and keep it to mathematics.

- Do not type anything that identifies you: your name, student ID, SCU email, phone number, address, or date of birth. The tutor does not need to know who you are in order to help with a gradient.
- Do not type anything about anyone else — a classmate’s name, their work, or their grade. That is their information to share, not yours.
- Keep out anything sensitive or unrelated to the course: health, finances, immigration or visa status, family matters, passwords, or anything you would not want quoted back to you.
- Before you paste or upload, look at what you are sending. A screenshot or a photo of your work can carry your name, a page header, or a browser tab in the background.
- The tutor is a study aid, not a counselor and not the instructor. If something personal is affecting your work in this course, bring it to office hours or to the university’s student services instead — those conversations belong with a person, and they stay confidential.

None of this is a reason to avoid the tutor. Ask it anything about the mathematics — what you tried, where you are stuck, why a step feels wrong. Just keep yourself, and everyone else, out of the transcript.

Working with the tutor. Choose the problem you are working on from the drop-down at the top of the panel, then tell the tutor what you have tried so far — the more of your own thinking you show, the more useful its questions become. The tutor can nudge you toward a starting point, and it can point you to sections and worked examples in these notes to study.

If your answer is wrong, the tutor will tell you so and help you understand *why* it is wrong: which step breaks, what misconception is behind it, and what picture or small example will let you see the failure for yourself. Note that this requires showing your reasoning — a bare answer with no work, right or wrong, will simply get asked for the thinking behind it. And what the tutor will not do is tell you how to correct an error — no fixed-up step, no “instead, you should...” Finding the right path once you understand why the old one fails is your part of the work.

Two models. Next to the problem picker is a “Model” drop-down offering *Careful* and *Quick*. Both are Gemini Flash models. *Careful* is the default: it thinks before it answers, so it takes a few seconds longer and is better at staying in the role of a tutor rather than a solutions manual. *Quick* is lighter and replies almost instantly. Either will coach you and neither will hand you an answer; switching between them costs nothing and does not disturb your conversation or your hint count. Try both — and tell me which you prefer.

Hints are limited on purpose. The point of the tutor is to get you unstuck, not to walk you to the answer, so hints are deliberately scarce. The exact rules:

- You get at most *three hints per problem, total* — the budget never resets. The panel always shows how many you have left on the problem you selected (“Hints left: 2/3”).
- *Every* reply from the tutor costs a hint — there are no free ones. Lead with what you have tried, so each reply can meet your actual thinking instead of spending itself asking for it.
- Restarting the conversation, reloading the page, or switching problems does not refill hints. Spend them thoughtfully — full solutions are posted after the due date.
- When your hints for a problem run out, bring your work to office hours — working through it together is exactly what office hours are for.

Tell me how it is going. Under every hint is a single question — “Was this hint helpful?” — with a thumb up and a thumb down. One tap is the whole thing, and it is the most useful second you can spend on this course: it tells me which hints land and which ones miss, hint by hint. Nothing but your answer and which problem you were on is sent.

For anything longer there is the “feedback” button at the top of the panel, which sends me a short note about how the tutor is working for you: whether it helped, and — in a box you can write as much as you like in — what would make it better. Tapping the thumb down offers it too. All of it is anonymous, it has nothing to do with your grade, and the tutor itself never sees it. If you want me to see the exchange you just had, there is a box to attach it; it is off unless you tick it. The tutor is new, and what you write is how it gets better.

Finally, know what the tutor will and will not do. It will not state answers, complete a solution, or turn a wrong answer into a right one — no matter how you ask. There is exactly one way to get an answer confirmed: state your conclusion *together with your own justification*. If your reasoning is sound and complete, the tutor will tell you that you have it; if there is a gap, it will point at the gap and let you close it. A bare answer, right or wrong, earns nothing.

4.1 Assignment 1

These problems exercise the hyperbolic identities, the derivatives of the hyperbolic functions and the inverse hyperbolic functions. If you would like to review the material first, see [Hyperbolic Functions](#).

1. Show that $\tanh^2(x) + \operatorname{sech}^2(x) = 1$, for all real numbers x .

Solution. We write both terms over the common denominator $\cosh^2(x)$ and use the identity $\cosh^2(x) - \sinh^2(x) = 1$, i.e. $\sinh^2(x) + 1 = \cosh^2(x)$:

$$\begin{aligned}\tanh^2(x) + \operatorname{sech}^2(x) &= \frac{\sinh^2(x)}{\cosh^2(x)} + \frac{1}{\cosh^2(x)} \\ &= \frac{\sinh^2(x) + 1}{\cosh^2(x)} = \frac{\cosh^2(x)}{\cosh^2(x)} = 1.\end{aligned}$$

Note that this is valid for every real x , since $\cosh(x) \geq 1 > 0$ and so the denominator is never zero.

2. Compute $\frac{d}{dx} 3 \sinh\left(\frac{3}{x}\right)$.

Solution. We use $\frac{d}{dx} \sinh(u) = \cosh(u) \frac{du}{dx}$ with $u = \frac{3}{x} = 3x^{-1}$, so $\frac{du}{dx} = -3x^{-2} = -\frac{3}{x^2}$. Hence

$$\frac{d}{dx} 3 \sinh\left(\frac{3}{x}\right) = 3 \cosh\left(\frac{3}{x}\right) \left(-\frac{3}{x^2}\right) = -\frac{9}{x^2} \cosh\left(\frac{3}{x}\right).$$

3. Simplify $\cosh(\ln 2)$.

Solution. Using the definition $\cosh x = \frac{e^x + e^{-x}}{2}$ together with $e^{\ln 2} = 2$ and $e^{-\ln 2} = \frac{1}{e^{\ln 2}} = \frac{1}{2}$,

$$\cosh(\ln 2) = \frac{e^{\ln 2} + e^{-\ln 2}}{2} = \frac{2 + \frac{1}{2}}{2} = \frac{\frac{5}{2}}{2} = \frac{5}{4}.$$

4. Solve the equation $5 \sinh(x) - 3 \cosh(x) = 3$ for x .

Solution. We first replace the hyperbolic functions by their definitions:

$$\begin{aligned}5 \sinh(x) - 3 \cosh(x) &= 3 \\ \Rightarrow \frac{5(e^x - e^{-x})}{2} - \frac{3(e^x + e^{-x})}{2} &= 3 \\ \Rightarrow e^x - 4e^{-x} &= 3.\end{aligned}$$

Multiplying through by e^x and writing $u = e^x$ turns this into a quadratic equation:

$$\begin{aligned}u^2 - 3u - 4 &= 0 \\ \Rightarrow (u - 4)(u + 1) &= 0.\end{aligned}$$

Since $u = e^x$ is positive, the root $u = -1$ is impossible, and only $u = 4$ survives. Therefore

$$e^x = 4 \quad \Rightarrow \quad x = \ln 4 = 2 \ln 2.$$

5. Following the method of [Example 1.4.2](#), show that the inverse hyperbolic tangent is given by

$$\tanh^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \quad -1 < x < 1.$$

Solution. Unlike $\cosh$, the function $\tanh$ is increasing on all of $\mathbb{R}$, so no restriction of its domain is needed. Set $y = \tanh(x)$ and solve for x . Multiplying the numerator and the denominator by e^x gives

$$y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot \frac{e^x}{e^x} = \frac{e^{2x} - 1}{e^{2x} + 1}.$$

Writing $u = e^{2x}$ and clearing the denominator,

$$y(u + 1) = u - 1$$

$$\begin{aligned}\Rightarrow y + 1 &= u - yu = u(1 - y) \\ \Rightarrow u &= \frac{1 + y}{1 - y}.\end{aligned}$$

Therefore $e^{2x} = \frac{1 + y}{1 - y}$, and taking the natural logarithm gives $2x = \ln\left(\frac{1 + y}{1 - y}\right)$. Interchanging the names of the two variables, so that $y = \tanh^{-1}(x)$ is the inverse function,

$$y = \tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right).$$

The range of $\tanh$ is the interval $(-1, 1)$, which is exactly the set of x for which $\frac{1+x}{1-x}$ is positive, so this is the domain of $\tanh^{-1}$.

6. Use the substitution $x = \tanh(t)$ to calculate

$$\int \frac{dx}{1 - x^2}, \quad -1 < x < 1,$$

and then use [Worksheet Exercise 4.1.5](#) to write your answer with a logarithm. *Hint:* $\frac{d}{dt} \tanh(t) = \operatorname{sech}^2(t)$, and $1 - \tanh^2(t) = \operatorname{sech}^2(t)$.

Solution. With $x = \tanh(t)$ we have $dx = \operatorname{sech}^2(t) dt$, and the denominator collapses by the identity in the hint:

$$\begin{aligned}\int \frac{dx}{1 - x^2} &= \int \frac{\operatorname{sech}^2(t)}{1 - \tanh^2(t)} dt \\ &= \int \frac{\operatorname{sech}^2(t)}{\operatorname{sech}^2(t)} dt \\ &= \int dt = t + C.\end{aligned}$$

Since $x = \tanh(t)$ means $t = \tanh^{-1}(x)$, and using [Worksheet Exercise 4.1.5](#) to rewrite the inverse function,

$$\begin{aligned}\int \frac{dx}{1 - x^2} &= \tanh^{-1}(x) + C \\ &= \frac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right) + C.\end{aligned}$$

This agrees with the answer obtained by partial fractions, since $\frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) = \frac{1}{2} \ln(1 + x) - \frac{1}{2} \ln(1 - x)$.

4.2 Assignment 2

These problems work with sequences, finite sums and geometric series, ask whether a series converges and what it sums to, and then use the Maclaurin series of the standard functions to recognise a sum. If you would like to review the material first, see [Infinite Series](#) and [Taylor and Maclaurin Series](#).

1. The terms of a geometric series $a + ar + ar^2 + \dots + ar^{n-1} + \dots$ satisfy

$$a_5 + a_7 = 1500 \quad \text{and} \quad a_{11} + a_{13} = 187500,$$

where $a_n = ar^{n-1}$ denotes the n th term. Find $a_1 + a_2 + a_3$.

Solution. Using $a_n = ar^{n-1}$,

$$\begin{aligned} a_5 + a_7 &= ar^4 + ar^6 = ar^4(1 + r^2) = 1500, \\ a_{11} + a_{13} &= ar^{10} + ar^{12} = ar^{10}(1 + r^2) = 187500. \end{aligned}$$

Dividing the two equations,

$$\frac{a_{11} + a_{13}}{a_5 + a_7} = \frac{ar^{10}(1 + r^2)}{ar^4(1 + r^2)} = \frac{187500}{1500} \Rightarrow r^6 = 125 = 5^3 \Rightarrow r^2 = 5.$$

Substituting back,

$$ar^4(1 + r^2) = 1500 \Rightarrow a(25)(1 + 5) = 1500 \Rightarrow a = \frac{1500}{150} = 10.$$

Since $r^2 = 5$, we have $r = \pm\sqrt{5}$, and therefore

$$\begin{aligned} a_1 + a_2 + a_3 &= a + ar + ar^2 = a(1 + r + r^2) \\ &= 10(1 \pm \sqrt{5} + 5) = 10(6 \pm \sqrt{5}). \end{aligned}$$

2. What is the sum of all multiples of 7 or 11 less than 1000?

Solution. Note that we need to subtract the multiples of 77, since they have been added twice:

$$\begin{aligned} &\underbrace{(7 + 14 + 21 + \dots + 77 + \dots + 994)}_{\text{multiples of 7}} \\ &+ \underbrace{(11 + 22 + \dots + 77 + \dots + 990)}_{\text{multiples of 11}} \\ &- \underbrace{(77 + 154 + \dots + 924)}_{\text{multiples of 77}} \\ &= 7(1 + 2 + 3 + \dots + 142) + 11(1 + 2 + 3 + \dots + 90) \\ &\quad - 77(1 + 2 + \dots + 12) \end{aligned}$$

$$\text{Using } \sum_{i=1}^n i = \frac{n(n+1)}{2}:$$

$$\begin{aligned} &= 7 \frac{(142)(143)}{2} + 11 \frac{(90)(91)}{2} - 77 \frac{(12)(13)}{2} \\ &= 7(71)(143) + 11(45)(91) - 77(6)(13) \\ &= (7)(71)(13)(11) + (11)(13)(7)(45) - (7)(11)(6)(13) \\ &= (7)(11)(13)(71 + 45 - 6) = (7)(11)(13)(110) = 110110, \end{aligned}$$

where we used $143 = (13)(11)$ and $91 = (13)(7)$.

3. What can we conclude by applying the n th term test in the following series?

A. $\sum_{n=1}^{\infty} \frac{(5n+3)^{2n}}{(7n^2+2)^n}$

B. $\sum_{n=1}^{\infty} \frac{2\ln(n)}{3n}$

Solution.

A.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{(5n+3)^{2n}}{(7n^2+2)^n} &= \lim_{n \rightarrow \infty} \left(\frac{(5n+3)^2}{7n^2+2} \right)^n \\ &= \lim_{n \rightarrow \infty} \left(\frac{25n^2+30n+9}{7n^2+2} \right)^n \\ &= \lim_{n \rightarrow \infty} \left(\frac{25}{7} \right)^n = \infty \Rightarrow \text{the series diverges.} \end{aligned}$$

B.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2\ln(n)}{3n} &\stackrel{H}{=} \frac{2}{3} \lim_{n \rightarrow \infty} \frac{1/n}{1} = 0 \\ &\Rightarrow \text{the } n\text{th Term Test is inconclusive.} \end{aligned}$$

4. Evaluate the following sums or show that they diverge.

A. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$

B. $\sum_{n=0}^{\infty} e^{-2n}$

C. $\sum_{n=0}^{\infty} e^{2n}$

D. $\sum_{n=1}^{\infty} \left(1 + \frac{4}{n} \right)^n$

E. $\sum_{n=2}^{\infty} \frac{2}{3^n}$

Solution.

A. We use partial fractions:

$$\begin{aligned} \frac{1}{n(n+1)(n+2)} &= \frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+2} \\ &= \frac{A(n+1)(n+2) + Bn(n+2) + Cn(n+1)}{n(n+1)(n+2)}. \end{aligned}$$

Setting $n = 0$ gives $2A = 1$, so $A = \frac{1}{2}$; setting $n = -1$ gives $-B = 1$, so $B = -1$; and setting $n = -2$ gives $2C = 1$, so $C = \frac{1}{2}$. Hence

$$\begin{aligned} &\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} \\ &= \sum_{n=1}^{\infty} \frac{1}{2} \left(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2} \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) - \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \\
&= \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) - \frac{1}{2} \sum_{n=2}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\
&= \frac{1}{2} \left(1 - \frac{1}{2} \right) \\
&\quad + \underbrace{\frac{1}{2} \sum_{n=2}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) - \frac{1}{2} \sum_{n=2}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)}_{\text{zero}} \\
&= \frac{1}{4}.
\end{aligned}$$

B. This is a geometric series with $a = 1$ and $r = \frac{1}{e^2}$:

$$\begin{aligned}
\sum_{n=0}^{\infty} e^{-2n} &= \sum_{n=0}^{\infty} \left(\frac{1}{e^2} \right)^n = \sum_{n=1}^{\infty} \left(\frac{1}{e^2} \right)^{n-1} \\
&= \frac{1}{1 - \frac{1}{e^2}} = \frac{e^2}{e^2 - 1}.
\end{aligned}$$

C. $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} e^{2n} = \infty$, so by the n th Term Test the series diverges.

D. Let $L = \lim_{n \rightarrow \infty} \left(1 + \frac{4}{n} \right)^n$. Then

$$\begin{aligned}
\ln L &= \lim_{n \rightarrow \infty} n \ln \left(1 + \frac{4}{n} \right) = \lim_{n \rightarrow \infty} \frac{\ln \left(1 + \frac{4}{n} \right)}{\frac{1}{n}} \\
&\stackrel{\text{H}}{=} \lim_{n \rightarrow \infty} \frac{\frac{-4/n^2}{1 + 4/n}}{-\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{4}{1 + \frac{4}{n}} = 4.
\end{aligned}$$

Hence $L = e^4 \neq 0$, so by the n th Term Test the series diverges.

E.

$$\begin{aligned}
\sum_{n=2}^{\infty} \frac{2}{3^n} &= \sum_{n=2}^{\infty} 2 \left(\frac{1}{3} \right)^n \\
&= \frac{2}{1 - \frac{1}{3}} - \left(2 + \frac{2}{3} \right) = 3 - \frac{8}{3} = \frac{1}{3},
\end{aligned}$$

where the terms for $n = 0$ and $n = 1$ need to be subtracted.

5. Answer questions A and B below for the following infinite series:

$$\sum_{k=1}^{\infty} \left(\frac{6}{k^2} - \frac{6}{(k+1)^2} \right).$$

A. Does the n th-term test apply? Remember to fully justify your answer.

B. Evaluate the series or show that it diverges.

Solution.

A. We compute the limit of the terms:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{6}{n^2} - \frac{6}{(n+1)^2} = \lim_{n \rightarrow \infty} \frac{6(n+1)^2 - 6n^2}{n^2(n+1)^2}$$

$$= \lim_{n \rightarrow \infty} \frac{12n + 6}{n^2(n+1)^2} = \lim_{n \rightarrow \infty} \frac{12n}{n^4} = 0.$$

Hence the n th-term test does *not* apply: since the limit of the terms is 0, the test is inconclusive.

B. The series telescopes. The partial sum is

$$\begin{aligned} s_n &= \left(\frac{6}{1^2} - \frac{6}{2^2} \right) + \left(\frac{6}{2^2} - \frac{6}{3^2} \right) + \cdots \\ &\quad + \left(\frac{6}{(n-1)^2} - \frac{6}{n^2} \right) + \left(\frac{6}{n^2} - \frac{6}{(n+1)^2} \right) \\ &= 6 - \frac{6}{(n+1)^2}, \end{aligned}$$

and therefore

$$\sum_{k=1}^{\infty} \left(\frac{6}{k^2} - \frac{6}{(k+1)^2} \right) = \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} 6 - \frac{6}{(n+1)^2} = 6 - 0 = 6.$$

6. Express 8.818181818... as a rational number, i.e. in the form $\frac{p}{q}$, where p and q are positive integers with no common factors.

Solution. We write the repeating decimal as a geometric series:

$$8.818181818\ldots = 8 + \frac{81}{100} + \frac{81}{10000} + \cdots = 8 + 81 \sum_{n=1}^{\infty} \left(\frac{1}{100} \right)^n.$$

Using the geometric series with $a = \frac{1}{100}$ and $r = \frac{1}{100}$,

$$8 + 81 \left(\frac{\frac{1}{100}}{1 - \frac{1}{100}} \right) = 8 + \frac{81}{99} = 8 + \frac{9}{11} = \frac{97}{11}.$$

Hence $p = 97$ and $q = 11$.

7. Find the sum of the convergent series $\sum_{n=0}^{\infty} \frac{(-25)^n}{(2n)!}$.

Solution. Recall $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$. With $x = 5$,

$$\begin{aligned} \cos(5) &= \sum_{n=0}^{\infty} \frac{(-1)^n 5^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n 25^n}{(2n)!} \\ &= \sum_{n=0}^{\infty} \frac{(-25)^n}{(2n)!}, \end{aligned}$$

so the sum is $\cos(5)$.

8. Find the sum of the series $\sum_{n=0}^{\infty} \frac{1}{n!}$.

Solution. The Maclaurin series of the exponential function is

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \text{valid for all } x \in \mathbb{R}.$$

Setting $x = 1$ gives

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e^1 = e.$$

9. Find the sum of the series

$$5 + 5^2 + \frac{5^3}{2!} + \frac{5^4}{3!} + \cdots + \frac{5^{n+1}}{n!} + \cdots$$

Solution. The general term is $\frac{5^{n+1}}{n!}$ starting at $n = 0$. Let us check the first few: at $n = 0$ we get $5^1/0! = 5$; at $n = 1$ we get $5^2/1! = 5^2$; at $n = 2$ we get $5^3/2!$. These match, so the series is

$$\sum_{n=0}^{\infty} \frac{5^{n+1}}{n!}.$$

Now pull out one factor of 5, which does not depend on n :

$$\sum_{n=0}^{\infty} \frac{5^{n+1}}{n!} = \sum_{n=0}^{\infty} \frac{5 \cdot 5^n}{n!} = 5 \sum_{n=0}^{\infty} \frac{5^n}{n!}.$$

The remaining series is the Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ evaluated at $x = 5$, which converges for every x . Therefore

$$5 + 5^2 + \frac{5^3}{2!} + \cdots = 5e^5.$$

4.3 Assignment 3

These problems bound the error in a Taylor approximation with the Remainder Theorem and use Taylor series to compute limits and integrals. Problems marked with a star (*) are for the interested reader: they do not need to be handed in and will not be examined. If you would like to review the material first, see [Taylor Series and its convergence](#).

1. For approximately what values of x can we replace $\sin(x)$ by $x - \frac{x^3}{6}$ with an error of magnitude no greater than 4×10^{-3} ?

Solution. Here $a = 0$, and the cubic $x - \frac{x^3}{6}$ is $T_3(x)$. But the Maclaurin series of $\sin$ has no x^4 term, so the same polynomial is also the fourth order Taylor polynomial:

$$\begin{aligned} T_4(x) &= x - \frac{x^3}{3!} + \frac{f^{(4)}(0)}{4!}x^4 \\ &= x - \frac{x^3}{3!} = T_3(x), \quad \text{since } f^{(4)}(0) = \sin(0) = 0. \end{aligned}$$

We should therefore take $n = 4$, not $n = 3$: both choices describe the very same approximation, but the remainder R_4 is one order smaller than R_3 and so gives the sharper estimate.

The derivatives of $f(x) = \sin(x)$ are

$$\begin{aligned} f'(x) &= \cos(x), & f''(x) &= -\sin(x), \\ f^{(3)}(x) &= -\cos(x), & f^{(4)}(x) &= \sin(x), \\ f^{(5)}(x) &= \cos(x), \end{aligned}$$

and therefore $|f^{(5)}(c)| = |\cos(c)| \leq 1$, so we can take $M = 1$.

By the Remainder Theorem with $n = 4$,

$$\text{Error} = |R_4(x)| \leq M \frac{|x - a|^5}{5!} = \frac{|x|^5}{5!} \leq 4 \times 10^{-3},$$

which gives

$$\begin{aligned} |x| &\leq \sqrt[5]{(5!)(0.004)} = \sqrt[5]{0.48} \approx 0.863, \\ \Rightarrow \quad x &\in [-0.863, 0.863]. \end{aligned}$$

See [Figure 4.3.1](#).

For comparison, stopping at $n = 3$ would have given $|x| \leq \sqrt[4]{(4!)(0.004)} \approx 0.556$, a needlessly small interval for exactly the same polynomial.

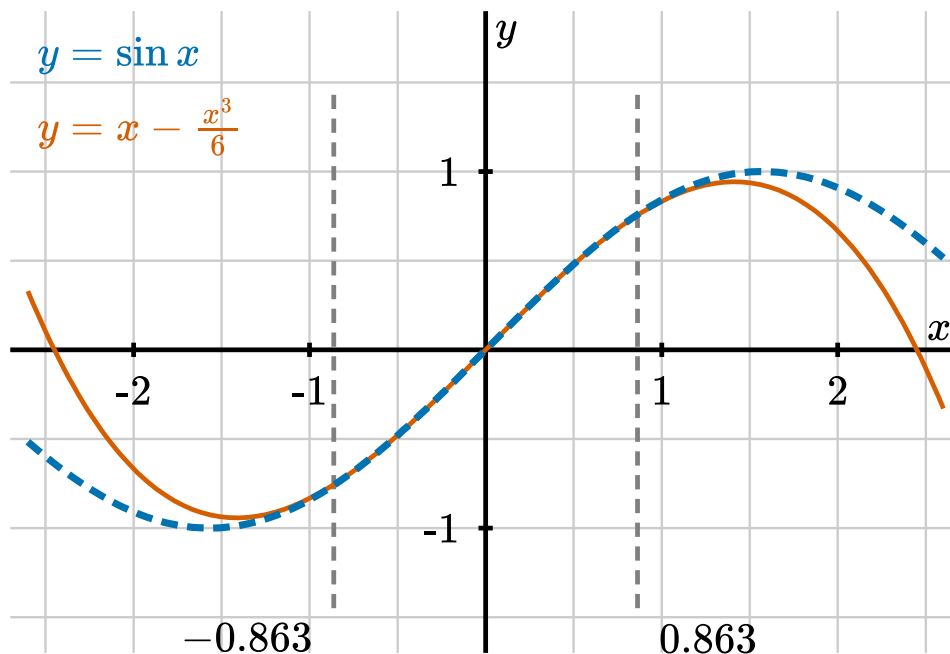


Figure 4.3.1 The function $y = \sin x$ and the cubic $y = x - \frac{x^3}{6}$. Between the dashed lines $x = \pm 0.863$ the two graphs differ by at most 4×10^{-3} .

2. Use the remainder estimation theorem to estimate the maximum error when approximating $f(x) = e^x$ by $T_2(x) = 1 + x + \frac{x^2}{2}$ on the interval $\left[-\frac{5}{6}, \frac{5}{6}\right]$.

Solution. From the remainder theorem, we know that

$$\text{Error} = |R_2(x)| = \left| \frac{f^{(3)}(c)}{3!} x^3 \right| \leq \frac{e^c}{3!} |x|^3.$$

Also, since $c \in \left[-\frac{5}{6}, \frac{5}{6}\right]$, we have $|c| \leq \frac{5}{6}$ and hence $e^c \leq e^{5/6}$:

$$\text{Error} = |R_2(x)| \leq \frac{e^{5/6}}{3!} \left(\frac{5}{6}\right)^3.$$

We know that $e^{5/6} < e^1 < 3$, hence

$$\text{Error} = |R_2(x)| \leq \frac{e^{5/6}}{3!} \left(\frac{5}{6}\right)^3 < \frac{3}{3!} \left(\frac{5}{6}\right)^3 \approx 0.289.$$

Hence, an estimate for the upper bound of the error is $\frac{3}{3!} \left(\frac{5}{6}\right)^3 \approx 0.289$. See [Figure 4.3.2](#).

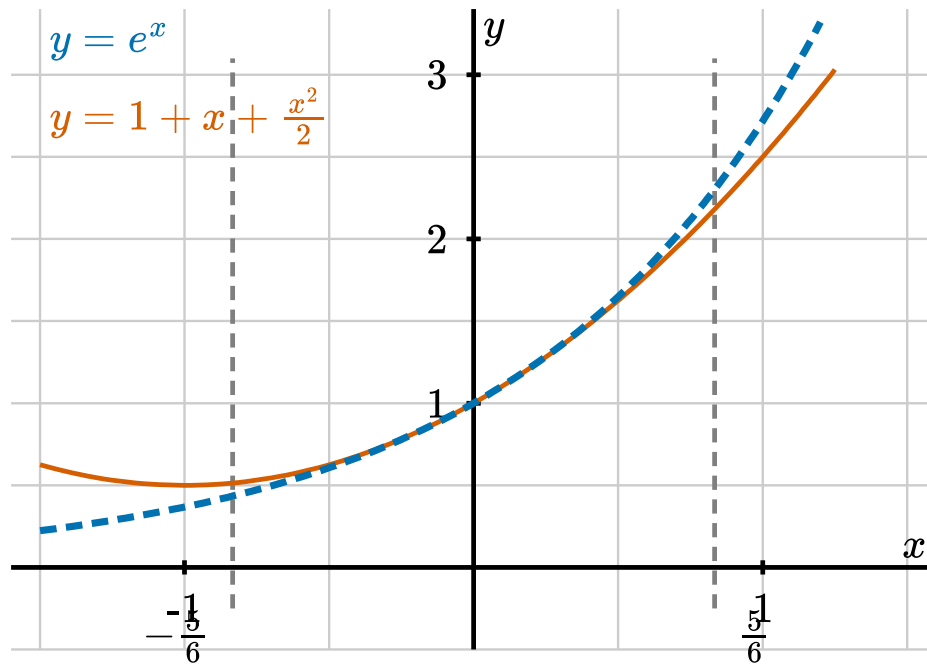


Figure 4.3.2 The function $y = e^x$ and $y = T_2(x) = 1 + x + \frac{x^2}{2}$ on the interval $[-\frac{5}{6}, \frac{5}{6}]$, marked by the dashed lines.

3. Consider the function $f(x) = \ln(1 + 2x)$.

- A. Find the second order Taylor polynomial $T_2(x)$ centered at 0.
- B. Estimate the maximum error when approximating $f(x) = \ln(1 + 2x)$ with the second order Taylor polynomial $T_2(x)$ centered at 0, on the interval $[-0.25, 0.25]$.

Solution.

- A. We compute the derivatives at 0:

$$\begin{aligned} f'(x) &= \frac{2}{1+2x} = 2(1+2x)^{-1}, & f'(0) &= 2, \\ f''(x) &= -4(1+2x)^{-2}, & f''(0) &= -4. \end{aligned}$$

Hence

$$T_2(x) = f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2 = 0 + 2x - 2x^2,$$

i.e. $T_2(x) = 2x - 2x^2$.

- B. We have $f^{(3)}(x) = 16(1+2x)^{-3}$ and $-\frac{1}{4} \leq x \leq \frac{1}{4}$. Note that $f^{(4)}(x) = -96(1+2x)^{-4}$ is negative, which means $f^{(3)}$ is a decreasing function, hence at $x = -\frac{1}{4}$ the value $f^{(3)}(c)$ is maximized, i.e.,

$$f^{(3)}(c) \leq f^{(3)}(-0.25) = 16 \left(1 - \frac{1}{2}\right)^{-3} = 128.$$

Also

$$-\frac{1}{4} \leq x \leq \frac{1}{4} \Rightarrow |x^3| \leq \left(\frac{1}{4}\right)^3 = \frac{1}{4^3} = \frac{1}{64},$$

and therefore

$$\text{Error} = |R_2(x)| \leq \left| \frac{f^{(3)}(c)}{3!} x^3 \right| \leq \frac{1}{3!} \cdot \frac{1}{64} \cdot 128 = \frac{1}{3}.$$

This means that the error in estimating $f(x) = \ln(1+2x)$ for $-\frac{1}{4} \leq x \leq \frac{1}{4}$ by $T_2(x) = 2x - 2x^2$ is definitely smaller than $\frac{1}{3}$. See Figure 4.3.3.

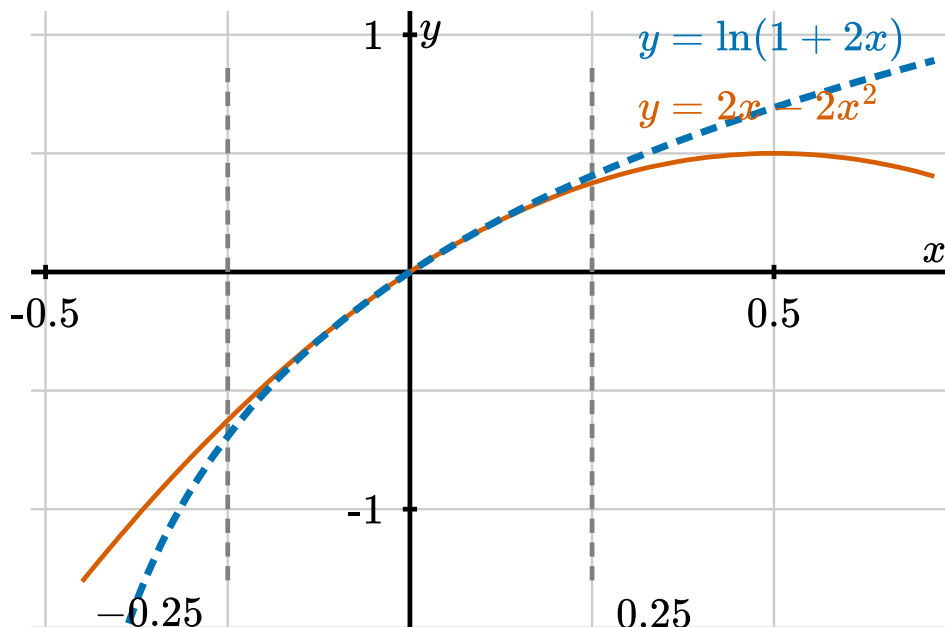


Figure 4.3.3 The function $y = \ln(1 + 2x)$ and $y = T_2(x) = 2x - 2x^2$. On the interval $[-0.25, 0.25]$, marked by the dashed lines, the two graphs are nearly identical.

4. Let

$$f(x) = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots$$

Find the exact value of $\int_0^{1/13} f(x) dx$.

Solution. First we identify f . Its general term is $(n+1)x^n$, so

$$f(x) = \sum_{n=0}^{\infty} (n+1)x^n.$$

This is exactly what you get by differentiating the geometric series term by term:

$$\begin{aligned} \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1, \\ \frac{d}{dx} \left(\frac{1}{1-x} \right) &= \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} = 1 + 2x + 3x^2 + \cdots \end{aligned}$$

So $f(x) = \frac{1}{(1-x)^2}$ for $|x| < 1$. Since $\frac{1}{13} < 1$, the whole interval of integration lies inside the interval of convergence.

Now we integrate. An antiderivative of $(1-x)^{-2}$ is $(1-x)^{-1}$, so

$$\begin{aligned} \int_0^{1/13} \frac{dx}{(1-x)^2} &= \left[\frac{1}{1-x} \right]_0^{1/13} \\ &= \frac{1}{1 - \frac{1}{13}} - \frac{1}{1-0} \\ &= \frac{1}{\frac{12}{13}} - 1 = \frac{13}{12} - 1 = \frac{1}{12}. \end{aligned}$$

5. Evaluate $\int \frac{\cos x - 1}{x} dx$ as an infinite series. Give the first three nonzero terms.

Solution. The integrand has no elementary antiderivative, so expand it as a series first. Starting from $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$, subtracting 1 kills the constant term, so every surviving power of x is even and at least 2. Dividing by x therefore leaves an honest power series:

$$\begin{aligned}\frac{\cos x - 1}{x} &= \frac{1}{x} \left(-\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) \\ &= -\frac{x}{2!} + \frac{x^3}{4!} - \frac{x^5}{6!} + \dots\end{aligned}$$

Integrating term by term,

$$\begin{aligned}\int \frac{\cos x - 1}{x} dx &= C - \frac{x^2}{2 \cdot 2!} + \frac{x^4}{4 \cdot 4!} - \frac{x^6}{6 \cdot 6!} + \dots \\ &= C - \frac{x^2}{4} + \frac{x^4}{96} - \frac{x^6}{4320} + \dots\end{aligned}$$

6. Use Taylor series to evaluate

$$\lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{x^2 e^x}.$$

Solution. The quotient has the indeterminate form $\frac{0}{0}$. Replace the numerator and the denominator by their Maclaurin series. Putting $2x$ in place of x in $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$,

$$\begin{aligned}\cos(2x) &= 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \dots = 1 - 2x^2 + \frac{2}{3}x^4 - \dots \\ \Rightarrow \cos(2x) - 1 &= -2x^2 + \frac{2}{3}x^4 - \dots\end{aligned}$$

and from $e^x = 1 + x + \frac{x^2}{2!} + \dots$,

$$x^2 e^x = x^2 \left(1 + x + \frac{x^2}{2!} + \dots \right) = x^2 + x^3 + \frac{x^4}{2} + \dots$$

Both vanish to exactly second order, so cancel the common factor of x^2 — legitimate, since the limit never uses the value at $x = 0$ itself:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\cos(2x) - 1}{x^2 e^x} &= \lim_{x \rightarrow 0} \frac{x^2 \left(-2 + \frac{2}{3}x^2 - \dots \right)}{x^2 \left(1 + x + \frac{x^2}{2!} + \dots \right)} \\ &= \lim_{x \rightarrow 0} \frac{-2 + \frac{2}{3}x^2 - \dots}{1 + x + \frac{x^2}{2!} + \dots} = \frac{-2}{1} = -2.\end{aligned}$$

Each remaining series is continuous at 0 with a nonzero value there, so only the two leading coefficients matter.

7. Show that

$$\int_0^1 \frac{\sin(\ln t)}{\ln t} dt = \frac{\pi}{4}.$$

(The integrand is continuous on $(0, 1)$. As $t \rightarrow 1^-$ we have $\ln t \rightarrow 0$, so the integrand tends to 1; as $t \rightarrow 0^+$ we have $\left| \frac{\sin(\ln t)}{\ln t} \right| \leq \frac{1}{|\ln t|} \rightarrow 0$. So the integrand is bounded and the integral exists. Its antiderivative is not elementary.)

Hint. Two facts from Calculus II, both proved by integration by parts¹: for every integer $k \geq 0$,

$$\begin{aligned}I_k &= \int_0^\infty x^k e^{-x} dx = k!, \\ J_k &= \int_0^1 (\ln t)^k dt = (-1)^k k!.\end{aligned}$$

Either substitute $t = e^{-x}$ and use [Example 2.3.13](#), or expand $\frac{\sin(\ln t)}{\ln t}$ as a series in $\ln t$ and integrate term by term with J_k .

Solution. *Solution 1 (substitution).* Let $t = e^{-x}$, so $x = -\ln t$ and $dt = -e^{-x} dx$. As t runs from 0 to 1, x runs from ∞ to 0. Since $\ln t = -x$,

$$\frac{\sin(\ln t)}{\ln t} = \frac{\sin(-x)}{-x} = \frac{\sin x}{x}.$$

Therefore

$$\begin{aligned} \int_0^1 \frac{\sin(\ln t)}{\ln t} dt &= \int_{\infty}^0 \frac{\sin x}{x} (-e^{-x}) dx \\ &= \int_0^{\infty} e^{-x} \frac{\sin x}{x} dx = \frac{\pi}{4} \end{aligned}$$

by [Example 2.3.13](#).

Solution 2 (directly with series). Substituting $u = \ln t$ into the series $\frac{\sin u}{u} = \sum_{n=0}^{\infty} \frac{(-1)^n u^{2n}}{(2n+1)!}$,

$$\frac{\sin(\ln t)}{\ln t} = \sum_{n=0}^{\infty} \frac{(-1)^n (\ln t)^{2n}}{(2n+1)!}.$$

Integrating term by term and using J_k with $k = 2n$ (so $(-1)^{2n} = 1$),

$$\begin{aligned} \int_0^1 \frac{\sin(\ln t)}{\ln t} dt &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (2n)! \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \arctan 1 = \frac{\pi}{4}. \end{aligned}$$

Justification. With $S_N(x) = \sum_{n=0}^N \frac{(-1)^n x^{2n}}{(2n+1)!}$, the Remainder Estimation Theorem gives $\left| \frac{\sin x}{x} - S_N(x) \right| \leq \frac{x^{2N+2}}{(2N+3)!}$, exactly as in [Example 2.3.13](#). Replacing x by $u = \ln t$,

$$\left| \frac{\sin(\ln t)}{\ln t} - S_N(\ln t) \right| \leq \frac{(\ln t)^{2N+2}}{(2N+3)!}.$$

Integrating over $[0, 1]$ with J_k , $k = 2N + 2$, bounds the error after $N + 1$ terms by

$$\frac{(2N+2)!}{(2N+3)!} = \frac{1}{2N+3} \rightarrow 0,$$

so the partial sums converge to the integral, and the integral equals $\pi/4$.

¹For I_k : $I_0 = \int_0^{\infty} e^{-x} dx = 1$, and for $k \geq 1$ integration by parts with $u = x^k$ and $dv = e^{-x} dx$ gives $I_k = \left[-x^k e^{-x} \right]_0^{\infty} + k I_{k-1} = k I_{k-1}$, since $x^k e^{-x} \rightarrow 0$ as $x \rightarrow \infty$; by induction $I_k = k!$. For J_k : $J_0 = 1$, and for $k \geq 1$ integration by parts with $u = (\ln t)^k$ and $dv = dt$, so $du = k(\ln t)^{k-1} \frac{dt}{t}$ and $v = t$, gives $J_k = \left[t(\ln t)^k \right]_0^1 - k J_{k-1} = -k J_{k-1}$. The bracket vanishes at $t = 1$ because $\ln 1 = 0$, and at $t = 0^+$ because $t(\ln t)^k \rightarrow 0$ (by L'Hôpital's rule). By induction, $J_k = (-k)(-k-1)\cdots(-1) J_0 = (-1)^k k!$.

4.4 Review Problems #1

These are the leftover problems from the first four topics of the course, collected here for review. They are not assigned, but they are fair game on an exam. Problems marked with a star (*) are the exception: those are for the interested reader and will not be examined. The topics they cover, with links to the notes:

- [Hyperbolic Functions](#)
- [Infinite Series](#)
- [Taylor and Maclaurin Series](#)
- [Taylor Series and its convergence](#)

1. **True or False.** For any real number x , we have $\cosh^2(x) + \sinh^2(x) = \cosh(2x)$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. True.

Solution. True.

First,

$$\cosh(2x) = \frac{1}{2} (e^{2x} + e^{-2x}).$$

Second,

$$\begin{aligned} \cosh^2(x) + \sinh^2(x) &= \frac{1}{4} (e^x + e^{-x})^2 \\ &\quad + \frac{1}{4} (e^x - e^{-x})^2 \\ &= \frac{1}{4} (2e^{2x} + 2e^{-2x}) \\ &= \frac{1}{2} (e^{2x} + e^{-2x}). \end{aligned}$$

Comparing the two results gives $\cosh^2(x) + \sinh^2(x) = \cosh(2x)$.

2. If $a, b, a+b$ and ab are positive numbers that form 4 consecutive terms in a geometric sequence, find a .

Solution. Since consecutive ratios in a geometric sequence are equal,

$$\frac{a}{b} = \frac{b}{a+b} = \frac{a+b}{ab}.$$

From the first equality,

$$\begin{aligned} \frac{a}{b} = \frac{b}{a+b} &\Rightarrow a^2 + ab = b^2 \Rightarrow b^2 - ab - a^2 = 0 \\ &\stackrel{\div a^2}{\Rightarrow} \left(\frac{b}{a}\right)^2 - \left(\frac{b}{a}\right) - 1 = 0 \\ \frac{b}{a} &= \frac{1 \pm \sqrt{1+4}}{2} \stackrel{\text{since } b/a > 0}{\Rightarrow} \frac{b}{a} = \frac{1 + \sqrt{5}}{2}. \quad (*) \end{aligned}$$

From the equality of the first and third fractions,

$$\begin{aligned} \frac{a}{b} = \frac{a+b}{ab} &\Rightarrow a^2 = a+b \stackrel{\div a}{\Rightarrow} a = 1 + \frac{b}{a} \\ &\stackrel{(*)}{=} 1 + \frac{1 + \sqrt{5}}{2} = \frac{3 + \sqrt{5}}{2}. \end{aligned}$$

3. Consider the sequence defined by $a_1 = 1$ and $a_{n+1} = a_n + 3n^2 + 3n + 1$. Find a formula for a_n and use it to compute a_{200} .

Solution. We compute the first few terms:

$$\begin{aligned} n = 1 : \quad a_2 &= a_1 + (3 + 3 + 1) = 1 + 7 = 8 \Rightarrow a_2 = 2^3 \\ n = 2 : \quad a_3 &= a_2 + (12 + 6 + 1) = 8 + 19 = 27 \Rightarrow a_3 = 3^3 \end{aligned}$$

We notice that $a_1 = 1^3$, $a_2 = 2^3$, and $a_3 = 3^3$, hence it seems that $a_n = n^3$.

Let us verify by computing $a_{n+1} - a_n$ using $a_n = n^3$:

$$\begin{aligned} a_{n+1} - a_n &= (n+1)^3 - n^3 \\ &= n^3 + 3n^2 + 3n + 1 - n^3 = 3n^2 + 3n + 1, \end{aligned}$$

which confirms that when $a_n = n^3$, then $a_{n+1} = a_n + 3n^2 + 3n + 1$. Hence,

$$a_{200} = (200)^3 = 8 \times 10^6.$$

4. What can we conclude by applying the n th term test to the series $\sum_{n=1}^{\infty} \frac{3n}{\sqrt{6n^2 + 4n + 5}}$?

Solution.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{3n}{\sqrt{6n^2 + 4n + 5}} &= \lim_{n \rightarrow \infty} \frac{3n}{\sqrt{6}n} = \sqrt{\frac{3}{2}} \neq 0 \\ &\Rightarrow \text{the series diverges.} \end{aligned}$$

5. Evaluate the following sums or show that they diverge.

A. $\sum_{n=0}^{10} e^{-2n}$

B. $\sum_{n=1}^{\infty} \frac{6}{(2n-1)(2n+1)}$

C. $\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2}$

D. $\sum_{n=1}^{\infty} \ln\left(\frac{n}{2n+1}\right)$

E. $\sum_{n=3}^{\infty} \frac{2^n}{3^n}$

Solution.

- A. This is a finite geometric sum. Using $\sum_{n=1}^N ar^{n-1} = a \frac{1-r^N}{1-r}$, which was proved when we first studied geometric sums,

$$\begin{aligned} \sum_{n=0}^{10} e^{-2n} &= \sum_{n=1}^{11} \left(\frac{1}{e^2}\right)^{n-1} \\ &= \frac{1 - (e^{-2})^{11}}{1 - e^{-2}} = \frac{1 - e^{-22}}{1 - e^{-2}}. \end{aligned}$$

- B. Partial fractions give

$$\frac{6}{(2n-1)(2n+1)} = \frac{A}{2n-1} + \frac{B}{2n+1} = \frac{A(2n+1) + B(2n-1)}{(2n-1)(2n+1)},$$

and setting $n = \frac{1}{2}$ gives $2A = 6$, $A = 3$, while $n = -\frac{1}{2}$ gives $-2B = 6$, $B = -3$. The sum telescopes:

$$\begin{aligned} s_k &= \left(\frac{1}{1} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \cdots \\ &\quad + \left(\frac{1}{2k-1} - \frac{1}{2k+1}\right) = 1 - \frac{1}{2k+1}, \\ \text{so } 3 \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} - \frac{1}{2n+1}\right) &= 3 \lim_{k \rightarrow \infty} s_k \\ &= 3 \lim_{k \rightarrow \infty} \left(1 - \frac{1}{2k+1}\right) = 3. \end{aligned}$$

C. Partial fractions (or direct observation) give

$$\frac{2n+1}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2},$$

so the sum telescopes:

$$\begin{aligned} s_k &= \sum_{n=1}^k \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) \\ &= \left(1 - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{9}\right) + \left(\frac{1}{9} - \frac{1}{16}\right) + \cdots \\ &\quad + \left(\frac{1}{k^2} - \frac{1}{(k+1)^2}\right) = 1 - \frac{1}{(k+1)^2} \\ \Rightarrow \sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2} &= \lim_{k \rightarrow \infty} \left(1 - \frac{1}{(k+1)^2}\right) = 1. \end{aligned}$$

D. $\lim_{n \rightarrow \infty} \ln\left(\frac{n}{2n+1}\right) = \ln\left(\frac{1}{2}\right) \neq 0$, so by the n th Term Test the series diverges.

E.

$$\begin{aligned} \sum_{n=3}^{\infty} \frac{2^n}{3^n} &= \sum_{n=4}^{\infty} \left(\frac{2}{3}\right)^{n-1} = \frac{1}{1 - \frac{2}{3}} - \left(\frac{4}{9} + \frac{2}{3} + 1\right) \\ &= 3 - \frac{4+6+9}{9} = 3 - \frac{19}{9} = \frac{27-19}{9} = \frac{8}{9}, \end{aligned}$$

where the subtracted terms correspond to $n = 3$, $n = 2$, and $n = 1$ of the shifted series.

6. The government has decided to give a \$1,000 tax rebate to each household in order to stimulate the economy. The government statistics say that each household will spend 80% of the rebate in goods and services. The businesses and individuals who benefited from that 80% will then spend 80% of what they received and so on. The result is called the multiplier effect. What is the total effect of the rebate on the economy?

Solution. The rounds of spending form a geometric series with $a = 1000$ and $r = 0.8$:

$$\begin{aligned} 1000 + 0.8(1000) + (0.8)^2(1000) + \cdots &= \sum_{n=1}^{\infty} 1000 (0.8)^{n-1} \\ &= \frac{1000}{1-0.8} = \frac{1000}{0.2} = 5000 \text{ dollars.} \end{aligned}$$

The successive rounds of spending are shown in [Figure 4.4.1](#).

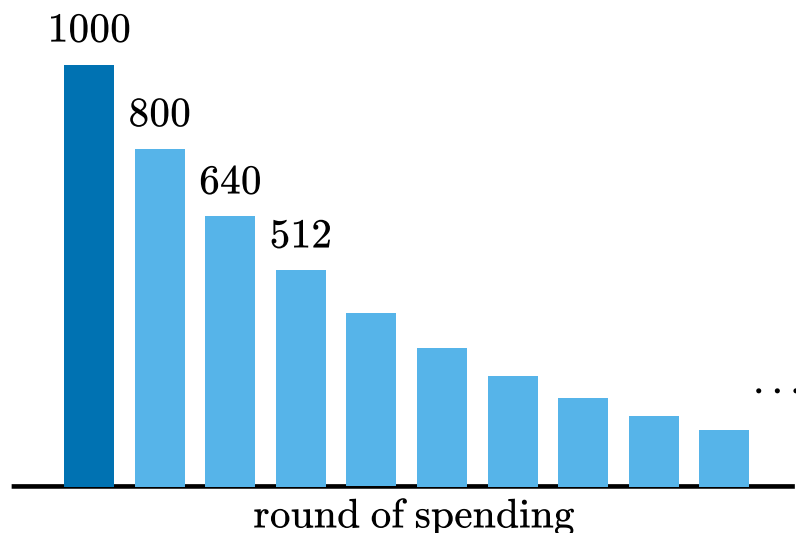


Figure 4.4.1 The multiplier effect. Each round of spending is 80% of the previous one, and the total of all the bars is $\frac{1000}{1-0.8} = 5000$ dollars.

7. Determine whether the series is convergent or divergent and if it is convergent calculate its sum.

A. $\sum_{n=0}^{\infty} \sin\left(\frac{1}{n+1}\right) - \sin\left(\frac{1}{n+2}\right)$

B. $\sum_{n=0}^{\infty} \frac{1+5^n}{3^n}$

Solution.

- A. The series telescopes. The partial sum is

$$\begin{aligned}
 s_k &= \sum_{n=0}^k \sin\left(\frac{1}{n+1}\right) - \sin\left(\frac{1}{n+2}\right) \\
 &= \sin(1) - \sin\left(\frac{1}{2}\right) + \sin\left(\frac{1}{2}\right) - \sin\left(\frac{1}{3}\right) \\
 &\quad + \sin\left(\frac{1}{3}\right) - \sin\left(\frac{1}{4}\right) + \cdots \\
 &\quad + \sin\left(\frac{1}{k+1}\right) - \sin\left(\frac{1}{k+2}\right) \\
 &= \sin(1) - \sin\left(\frac{1}{k+2}\right).
 \end{aligned}$$

Hence the series converges and

$$\begin{aligned}
 \sum_{n=0}^{\infty} \sin\left(\frac{1}{n+1}\right) - \sin\left(\frac{1}{n+2}\right) &= \lim_{k \rightarrow \infty} s_k \\
 &= \sin(1) - \sin(0) = \sin(1).
 \end{aligned}$$

- B. We split the series into two geometric series:

$$\sum_{n=0}^{\infty} \frac{1+5^n}{3^n} = \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n + \sum_{n=0}^{\infty} \left(\frac{5}{3}\right)^n.$$

Since $\lim_{n \rightarrow \infty} \left(\frac{5}{3}\right)^n = \infty$, by the n th term test we can conclude that the series diverges.

8. Determine whether the series converges, and if so find its sum:

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+5} \right)^n.$$

Solution. Note that

$$\begin{aligned} \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \left(1 + \frac{5}{n} \right)^{-n} \\ &= \lim_{n \rightarrow \infty} \left[\left(1 + \frac{5}{n} \right)^{\frac{n}{5}} \right]^{-5} = e^{-5} \neq 0, \end{aligned}$$

since $(1 + \frac{5}{n})^{n/5} \rightarrow e$. Indeed, setting $x = \frac{n}{5}$ and $L = \lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x$,

$$\begin{aligned} \ln L &= \lim_{x \rightarrow \infty} x \ln \left(1 + \frac{1}{x} \right) = \lim_{x \rightarrow \infty} \frac{\ln(1 + \frac{1}{x})}{\frac{1}{x}} \\ &\stackrel{\text{H}}{=} \lim_{x \rightarrow \infty} \frac{\frac{-1/x^2}{1 + 1/x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} = 1 \\ &\Rightarrow L = e. \end{aligned}$$

Therefore, the series diverges by the n th Term Test.

9. Consider the following series. Answer the following questions.

$$\sum_{n=0}^{\infty} \frac{(x+4)^n}{2^n}$$

A. Find the values of x for which the series converges.

B. Find the sum of the series for those values of x .

Solution.

A. We can write the series as a geometric series:

$$\sum_{n=0}^{\infty} \frac{(x+4)^n}{2^n} = \sum_{n=0}^{\infty} \left(\frac{x+4}{2} \right)^n = \sum_{n=0}^{\infty} \left(\frac{x}{2} + 2 \right)^n,$$

which converges when $\left| \frac{x}{2} + 2 \right| < 1$:

$$-1 < \frac{x}{2} + 2 < 1 \Rightarrow -3 < \frac{x}{2} < -1 \Rightarrow -6 < x < -2,$$

i.e. the series converges on the interval $(-6, -2)$.

B. When $x \in (-6, -2)$, the geometric series with $a = 1$ and $r = \frac{x+4}{2}$ gives

$$\sum_{n=0}^{\infty} \left(\frac{x+4}{2} \right)^n = \frac{1}{1 - \frac{x+4}{2}} = \frac{2}{2 - x - 4} = -\frac{2}{x+2}.$$

10. **True or False.** A geometric series $\sum_{n=1}^{\infty} ar^{n-1}$ converges if and only if $|r| \leq 1$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

As a counterexample consider $\sum_{n=1}^{\infty} 1^{n-1}$, which clearly diverges even though $|r| = 1$. The correct condition is $|r| < 1$.

11. Consider the series $\sum_{k=0}^{\infty} \frac{3^{k-1}}{5^k}$. Does the series converge? If so, what is the result?

Solution. We first factor out $\frac{1}{3}$ so that the powers match:

$$\sum_{k=0}^{\infty} \frac{3^{k-1}}{5^k} = \frac{1}{3} \sum_{k=0}^{\infty} \frac{3^k}{5^k} = \frac{1}{3} \sum_{k=0}^{\infty} \left(\frac{3}{5}\right)^k.$$

Splitting off the $k = 0$ term so that the sum starts at $k = 1$,

$$\frac{1}{3} \sum_{k=0}^{\infty} \left(\frac{3}{5}\right)^k = \frac{1}{3} \left[1 + \sum_{k=1}^{\infty} \left(\frac{3}{5}\right)^k \right].$$

The remaining sum is a geometric series with $a = \frac{3}{5}$ and $r = \frac{3}{5}$. Since $|r| = \frac{3}{5} < 1$, it converges and $\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}$, so

$$\frac{1}{3} \left[1 + \frac{\frac{3}{5}}{1 - \frac{3}{5}} \right] = \frac{1}{3} \left[1 + \frac{3}{2} \right] = \frac{1}{3} \cdot \frac{5}{2} = \frac{5}{6}.$$

12. Find the sum of all of the convergent series.

A. $\sum_{n=0}^{\infty} \frac{\sin(\frac{\pi}{2}n)}{3^{n+1}}$

B. $\sum_{n=1}^{\infty} \frac{-2}{n(n+1)}$

Solution.

- A. The factor $\sin(\frac{\pi}{2}n)$ cycles through $0, 1, 0, -1$ as $n = 0, 1, 2, 3, \dots$, so only the odd n contribute, with alternating signs. Writing $n = 2k + 1$,

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{\sin(\frac{\pi}{2}n)}{3^{n+1}} &= \frac{1}{3^2} - \frac{1}{3^4} + \frac{1}{3^6} - \dots \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{3^{2k+2}} = \frac{1}{9} \sum_{k=0}^{\infty} \left(-\frac{1}{9}\right)^k. \end{aligned}$$

This is a geometric series with $a = 1$ and $r = -\frac{1}{9}$, and $|r| < 1$, so

$$\frac{1}{9} \cdot \frac{1}{1 - (-\frac{1}{9})} = \frac{1}{9} \cdot \frac{1}{\frac{10}{9}} = \frac{1}{10}.$$

- B. By the constant multiple rule, $\sum_{n=1}^{\infty} \frac{-2}{n(n+1)} = -2 \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. The remaining series telescopes: since

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1},$$

$$\begin{aligned} s_n &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} \longrightarrow 1. \end{aligned}$$

Hence the sum is $-2(1) = -2$.

- 13. True or False.** If we manage to find the maximum value of $|f^{(n+1)}(c)|$ for c in the interval between a and x , then we can find the exact error in the Taylor polynomial approximation.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

The statement is incorrect. While finding the maximum value of $|f^{(n+1)}(c)|$ allows us to find an upper bound for the error, it does not give the exact error.

- 14.** For the following functions, give the Taylor *series* generated by the function at $x = 0$ AND give the values of x for which the Taylor series converges to the given function.

A. $f(x) = \cos(x)$

B. $g(x) = \frac{1}{1-x}$

C. $h(x) = x^2 e^{x^2}$

- D. Using the Taylor series you found for $h(x)$ in part C, find $h^{(12)}(0)$, i.e. the 12th order derivative of $h(x)$ at 0.

Solution.

- A. The derivatives of $\cos(x)$ at 0 cycle through $1, 0, -1, 0$, so only even powers survive and

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots,$$

which converges to $\cos(x)$ for all $x \in \mathbb{R}$.

- B. This is the geometric series with $a = 1$ and $r = x$:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots,$$

which converges to $\frac{1}{1-x}$ exactly when $|x| < 1$, i.e. for $-1 < x < 1$.

- C. We start from $e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!}$, valid for all u , and substitute $u = x^2$:

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}.$$

Multiplying by x^2 ,

$$h(x) = x^2 e^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n+2}}{n!} = x^2 + x^4 + \frac{x^6}{2!} + \frac{x^8}{3!} + \cdots,$$

which converges to $h(x)$ for all $x \in \mathbb{R}$.

- D. In the Taylor series $h(x) = \sum_{k=0}^{\infty} \frac{h^{(k)}(0)}{k!} x^k$, the coefficient of x^{12} is $\frac{h^{(12)}(0)}{12!}$. From part C, the term with x^{12} occurs when $2n + 2 = 12$, i.e. $n = 5$, and its coefficient is $\frac{1}{5!}$. Equating the two,

$$\frac{h^{(12)}(0)}{12!} = \frac{1}{5!} \Rightarrow h^{(12)}(0) = \frac{12!}{5!} = \frac{479001600}{120} = 3991680.$$

- 15.** Given $f(x) = \ln(1 - 2x)$.

- A. Find the 2nd order Maclaurin polynomial for $f(x)$.
- B. Use Taylor's Remainder Theorem to find an upper bound on the magnitude of the error if the 2nd order Maclaurin polynomial is used to approximate $f(x)$ for $|x| \leq \frac{1}{4}$.

Solution.

- A. We compute the derivatives at 0:

$$\begin{aligned} f(x) &= \ln(1 - 2x), & f(0) &= 0, \\ f'(x) &= \frac{-2}{1 - 2x} = -2(1 - 2x)^{-1}, & f'(0) &= -2, \\ f''(x) &= -4(1 - 2x)^{-2}, & f''(0) &= -4. \end{aligned}$$

Hence

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 = -2x - 2x^2.$$

- B. By Taylor's Remainder Theorem,

$$|R_2(x)| \leq M \frac{|x|^3}{3!},$$

where $M \geq |f^{(3)}(c)|$ for all c between 0 and x .

Here $f^{(3)}(x) = -16(1 - 2x)^{-3}$, so $|f^{(3)}(c)| = \frac{16}{|1 - 2c|^3}$. For $|c| \leq \frac{1}{4}$ we have $\frac{1}{2} \leq 1 - 2c \leq \frac{3}{2}$, and the fraction is largest when the denominator is smallest, i.e. at $c = \frac{1}{4}$:

$$|f^{(3)}(c)| \leq \frac{16}{(\frac{1}{2})^3} = 16 \cdot 8 = 128 = M.$$

Therefore, for $|x| \leq \frac{1}{4}$,

$$|R_2(x)| \leq 128 \cdot \frac{(\frac{1}{4})^3}{3!} = \frac{128}{64 \cdot 6} = \frac{1}{3}.$$

16. What function has Maclaurin series $x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!} + \frac{x^9}{8!} - \dots$?

Solution. Factoring x out of every term,

$$x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!} + \dots = x \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right),$$

and the bracket is exactly the Maclaurin series of $\cos(x)$. Hence the function is

$$x \cos(x).$$

17.

- A. Use a third Taylor polynomial at $x = 100$ to approximate $\sqrt{102}$.
- B. Give an upper bound for the error in using this approximation.

Solution.

- A. Let $f(x) = \sqrt{x} = x^{1/2}$ and $a = 100$. Then

$$\begin{aligned} f(x) &= x^{1/2}, & f(100) &= 10, \\ f'(x) &= \frac{1}{2}x^{-1/2}, & f'(100) &= \frac{1}{20}, \\ f''(x) &= -\frac{1}{4}x^{-3/2}, & f''(100) &= -\frac{1}{4000}, \end{aligned}$$

$$f'''(x) = \frac{3}{8}x^{-5/2}, \quad f'''(100) = \frac{3}{800000}.$$

Hence

$$T_3(x) = 10 + \frac{x-100}{20} - \frac{(x-100)^2}{8000} + \frac{(x-100)^3}{1600000},$$

and evaluating at $x = 102$ (so $x - 100 = 2$),

$$\begin{aligned} \sqrt{102} &\approx 10 + \frac{2}{20} - \frac{4}{8000} + \frac{8}{1600000} \\ &= 10 + \frac{1}{10} - \frac{1}{2000} + \frac{1}{200000}. \end{aligned}$$

B. By Taylor's Remainder Theorem, $|R_3(x)| \leq M \frac{|x-100|^4}{4!}$, where M bounds $|f^{(4)}|$ on $[100, 102]$. Since $f^{(4)}(x) = -\frac{15}{16}x^{-7/2}$ and $x^{-7/2}$ is decreasing, the largest value occurs at $x = 100$:

$$M = \frac{15}{16} \cdot 100^{-7/2} = \frac{15}{16 \cdot 10^7}.$$

Therefore

$$|R_3(102)| \leq \frac{15}{16 \cdot 10^7} \cdot \frac{2^4}{24} = \frac{15}{24 \cdot 10^7} = \frac{1}{16000000}.$$

18. Use Taylor series to evaluate

$$\lim_{x \rightarrow 0} \frac{\arctan x - x}{\sin x - x}.$$

Solution. Numerator and denominator both tend to 0, and each vanishes to third order, so l'Hôpital's rule would have to be applied three times. Replacing each by its Maclaurin series is quicker. From $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ and $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$, subtracting x removes the leading term of each:

$$\begin{aligned} \arctan x - x &= -\frac{x^3}{3} + \frac{x^5}{5} - \dots \\ \sin x - x &= -\frac{x^3}{3!} + \frac{x^5}{5!} - \dots \end{aligned}$$

Both series begin with x^3 . The limit never sees $x = 0$ itself, so we may cancel a factor of x^3 from top and bottom:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\arctan x - x}{\sin x - x} &= \lim_{x \rightarrow 0} \frac{-\frac{1}{3} + \frac{x^2}{5} - \dots}{-\frac{1}{3!} + \frac{x^2}{5!} - \dots} \\ &= \frac{\lim_{x \rightarrow 0} \left(-\frac{1}{3} + \frac{x^2}{5} - \dots \right)}{\lim_{x \rightarrow 0} \left(-\frac{1}{3!} + \frac{x^2}{5!} - \dots \right)} \\ &= \frac{-\frac{1}{3}}{-\frac{1}{3!}} = \frac{-\frac{1}{3}}{-\frac{1}{6}} = 2. \end{aligned}$$

Each of the two remaining series is a power series with a nonzero constant term, so it is continuous at 0 and the limit is just that constant term. Only the leading coefficients matter: the answer is the ratio $(-\frac{1}{3}) / (-\frac{1}{6})$.

4.5 Sample Past Exam 1

A sample past exam, with worked solutions. It was a 65-minute, closed-book exam worth 20 points: no aids of their own (formula sheets, notes, phones, tablets, smart watches or similar) were permitted, and students were asked to write their work in a neat and organized format and to fully justify every step. The formula sheet at the end of this page will be the last page of the exam booklet on the day of the exam. Try the exam under the same conditions before you look at the solutions. The topics it covers, with links to the notes:

- [Hyperbolic Functions](#)
- [Infinite Series](#)
- [Taylor and Maclaurin Series](#)
- [Taylor Series and its convergence](#)

1. **4 points.** Answer parts (a) and (b) for the following infinite series:

$$\sum_{k=1}^{\infty} (e^{1-k} - e^{-k}).$$

- (a) Does the n th-term test apply? Remember to fully justify your answer. (2 points)

Solution.

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} (e^{1-k} - e^{-k}) = 0,$$

so the n th-term test does *not* apply: a limit of 0 is inconclusive.

- (b) Evaluate the series or show that it diverges. (2 points)

Solution. The partial sums telescope:

$$\begin{aligned} S_n &= \underbrace{(e^0 - e^{-1})}_{k=1} + \underbrace{(e^{-1} - e^{-2})}_{k=2} + \underbrace{(e^{-2} - e^{-3})}_{k=3} \\ &\quad + \cdots + \underbrace{(e^{2-n} - e^{1-n})}_{k=n-1} + \underbrace{(e^{1-n} - e^{-n})}_{k=n} \\ &= e^0 - e^{-n} = 1 - \frac{1}{e^n}. \end{aligned}$$

Hence

$$\sum_{k=1}^{\infty} (e^{1-k} - e^{-k}) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{e^n}\right) = 1.$$

2. **5 points.** Consider the Taylor series for the function $f(x) = \sinh(x)$ about $x = 0$.

- (a) Find $p_5(x)$, i.e. the Taylor polynomial of order 5. (2 points)

Solution.

$$\begin{array}{ll} f(x) = \sinh(x), & f(0) = 0, \\ f'(x) = \cosh(x), & f'(0) = 1, \\ f''(x) = \sinh(x), & f''(0) = 0, \\ f^{(3)}(x) = \cosh(x), & f^{(3)}(0) = 1, \\ f^{(4)}(x) = \sinh(x), & f^{(4)}(0) = 0, \\ f^{(5)}(x) = \cosh(x), & f^{(5)}(0) = 1. \end{array}$$

Therefore

$$p_5(x) = f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2$$

$$\begin{aligned}
& + \cdots + \frac{f^{(5)}(0)}{5!}(x-0)^5 \\
& = x + \frac{x^3}{3!} + \frac{x^5}{5!}.
\end{aligned}$$

- (b) Express the Taylor series for the function $f(x) = \sinh(x)$ about $x = 0$ in sigma notation. (1 point)

Solution.

$$\sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

- (c) Assuming that $-2 \leq x \leq 2$, find an upper bound on the error in estimating $f(x)$ using $p_5(x)$. (2 points)

Note: an answer in the form $(e^5 + e^{-3}) \frac{3^{10}}{9!}$ suffices.

Solution. Collect the ingredients of the Remainder Estimation Theorem:

- (I) $-2 \leq x \leq 2$, so $|x| \leq 2$.
 (II) $n = 6$, since $p_5(x) = p_6(x)$ for $f(x) = \sinh(x)$ (the coefficient of x^6 is $f^{(6)}(0)/6! = 0$).
 (III) $a = 0$.

- (IV) $f^{(n+1)}(t) = f^{(7)}(t) = \cosh(t) = \frac{e^t + e^{-t}}{2}$, so for $-2 \leq t \leq 2$,

$$\left| f^{(7)}(t) \right| \leq \frac{e^2 + e^{-2}}{2}, \quad \text{and we take} \quad M = \frac{e^2 + e^{-2}}{2}.$$

Therefore

$$\text{Error} = |R_5(x)| = |R_6(x)| \leq \frac{M|x-0|^7}{7!} \leq \left(\frac{e^2 + e^{-2}}{2} \right) \frac{2^7}{7!},$$

which is the upper bound on the error.

3. 4 points. Consider the series

$$\sum_{k=1}^{\infty} \left(\frac{2}{x} \right)^k.$$

- (a) For which values of x does the series converge? (2 points)

Solution.

$$\sum_{k=1}^{\infty} \left(\frac{2}{x} \right)^k = \sum_{k=1}^{\infty} \left(\frac{2}{x} \right) \left(\frac{2}{x} \right)^{k-1}$$

is a geometric series with $r = \frac{2}{x}$ and $a = \frac{2}{x}$. It converges when

$$|r| < 1 \Rightarrow \left| \frac{2}{x} \right| < 1 \Rightarrow |x| > 2 \Rightarrow x \in (-\infty, -2) \cup (2, \infty).$$

- (b) Evaluate the series for the x values that you found in part (a). (2 points)

Solution. Since

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \quad \text{if } |r| < 1,$$

we get

$$\sum_{k=1}^{\infty} \left(\frac{2}{x} \right)^k = \frac{\frac{2}{x}}{1 - \frac{2}{x}} = \frac{2}{x-2}.$$

4. **3 points.** Compute the result of

$$\sum_{n=1}^{\infty} \frac{2^{(n+1)}}{3^n}.$$

Solution.

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{2^{n+1}}{3^n} &= 2 \sum_{n=1}^{\infty} \frac{2^n}{3^n} = 2 \sum_{n=1}^{\infty} \left(\frac{2}{3}\right) \left(\frac{2}{3}\right)^{n-1} \\ &= 2 \frac{\frac{2}{3}}{1 - \frac{2}{3}} = 2 \frac{\frac{2}{3}}{\frac{1}{3}} = 4. \end{aligned}$$

The last sum is a geometric series with $a = \frac{2}{3}$ and $r = \frac{2}{3}$, which converges to $a/(1-r)$.

5. **4 points.** Find the result of the following series or show that the series diverges.

(a) $-\ln(3) + \frac{(\ln(3))^2}{2} - \frac{(\ln(3))^3}{6} + \frac{(\ln(3))^4}{24} - \dots$ (2 points)

Solution. Add and subtract 1:

$$\begin{aligned} &-\ln(3) + \frac{(\ln(3))^2}{2} - \frac{(\ln(3))^3}{6} + \frac{(\ln(3))^4}{24} - \dots \\ &= -1 + \left[1 - \ln(3) + \frac{(\ln(3))^2}{2} - \frac{(\ln(3))^3}{6} + \frac{(\ln(3))^4}{24} - \dots \right]. \end{aligned}$$

The bracket is the Maclaurin series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

with $x = -\ln(3)$. Hence the series equals

$$-1 + e^{-\ln(3)} = -1 + \frac{1}{e^{\ln(3)}} = -1 + \frac{1}{3} = -\frac{2}{3}.$$

(b) $e + e^2 + e^4 + e^6 + \dots$ (2 points)

Solution.

$$e + e^2 + e^4 + e^6 + \dots = e + \sum_{n=1}^{\infty} e^{2n},$$

and $\sum_{n=1}^{\infty} e^{2n}$ is a geometric series with $r = e^2$. Hence it diverges, since $|r| > 1$.

Formula Sheet.

Taylor's Formula. If f has derivatives of all orders in an open interval I containing a , then for each positive integer n and for each x in I ,

$$\begin{aligned} f(x) &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 \\ &\quad + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x) \end{aligned}$$

where $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ and c is between a and x .

The Remainder Estimation Theorem. If there is a positive constant M such that $|f^{(n+1)}(c)| \leq M$ for

all c between x and a , inclusive, then the remainder term $R_n(x)$ in Taylor's Theorem satisfies the inequality

$$\text{Error} = |R_n(x)| \leq M \frac{|x - a|^{n+1}}{(n+1)!}.$$

If this condition holds for every n and the other conditions of Taylor's Theorem are satisfied by f , then the series converges to $f(x)$.

Maclaurin series.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad x \in (-1, 1)$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$